Linear Algebra and Its Applications

Chapter 3

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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3.1 Introduction to determinants

A determinant is a number attached to a square matrix. For a 2×2 matrix it is $ad - bc$. In a larger matrix, delete row i and column j to obtain the minor matrix A_{ij}; its cofactor is the *number* $C_{ij} = (-1)^{i+j} \det A_{ij}$. The signs form a checkerboard, beginning with plus in position $(1, 1)$.

A cofactor expansion multiplies every entry of one chosen row or column by its cofactor, then adds. A zero entry contributes zero. Row or column positions in each new minor must be counted anew: they are not the original positions. For numerical checks below all arithmetic was also performed exactly with rational-number matrix computations.

Practice problem 3.1.P1: A sparse fourth-order determinant

What we must show. Compute the determinant without expanding sixteen terms.

Plan. Column 3 has just one nonzero entry, 2 in row 1. Its sign is $(-1)^{1+3} = 1$.

$$D = 2 \begin{vmatrix} 0 & 3 & -4 \\ -5 & -8 & 3 \\ 0 & 5 & -6 \end{vmatrix}.$$

Expand that minor down its first column. The only nonzero entry is now -5 in position $(2, 1)$, whose cofactor sign is negative. Hence

$$D = 2[-(-5)] \begin{vmatrix} 3 & -4 \\ 5 & -6 \end{vmatrix} = 10[3(-6) - (-4)5] = 10(2) = 20.$$

Conclusion. The determinant is 20. Keeping track of the sign and the negative entry separately prevents a common sign error.

Exercise 3.1.1: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (3)(-1)^{1+1} \begin{vmatrix} 3 & 2 \\ 5 & -1 \end{vmatrix} + (4)(-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 0 & 5 \end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\det A_{11} = (3)(-1) - (2)(5) = -13$$
$$\det A_{13} = (2)(5) - (3)(0) = 10.$$

Check by expanding down column 2, as requested.

$$\det A = (3)(-1)^{2+2}\begin{vmatrix} 3 & 4 \\ 0 & -1 \end{vmatrix} + (5)(-1)^{3+2}\begin{vmatrix} 3 & 4 \\ 2 & 2 \end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\det A_{22} = (3)(-1) - (4)(0) = -3$$
$$\det A_{32} = (3)(2) - (4)(2) = -2.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 1$.

Exercise 3.1.2: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 0 & 5 & 1 \\ 4 & -3 & 0 \\ 2 & 4 & 1 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (5)(-1)^{1+2}\begin{vmatrix} 4 & 0 \\ 2 & 1 \end{vmatrix} + (1)(-1)^{1+3}\begin{vmatrix} 4 & -3 \\ 2 & 4 \end{vmatrix} = 2.$$

The 2×2 determinants used above are

$$\det A_{12} = (4)(1) - (0)(2) = 4$$
$$\det A_{13} = (4)(4) - (-3)(2) = 22.$$

Check by expanding down column 2, as requested.

$$\det A = (5)(-1)^{1+2}\begin{vmatrix} 4 & 0 \\ 2 & 1 \end{vmatrix} + (-3)(-1)^{2+2}\begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix} 0 & 1 \\ 4 & 0 \end{vmatrix} = 2.$$

The 2×2 determinants used above are

$$\det A_{12} = (4)(1) - (0)(2) = 4$$
$$\det A_{22} = (0)(1) - (1)(2) = -2$$
$$\det A_{32} = (0)(0) - (1)(4) = -4.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 2$.

Exercise 3.1.3: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 2 & -4 & 3 \\ 3 & 1 & 2 \\ 1 & 4 & -1 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (2)(-1)^{1+1}\begin{vmatrix} 1 & 2 \\ 4 & -1 \end{vmatrix} + (-4)(-1)^{1+2}\begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} + (3)(-1)^{1+3}\begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} = -5.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (1)(-1) - (2)(4) = -9 \\ \det A_{12} &= (3)(-1) - (2)(1) = -5 \\ \det A_{13} &= (3)(4) - (1)(1) = 11. \end{aligned}$$

Check by expanding down column 2, as requested.

$$\det A = (-4)(-1)^{1+2}\begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} + (1)(-1)^{2+2}\begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -5.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{12} &= (3)(-1) - (2)(1) = -5 \\ \det A_{22} &= (2)(-1) - (3)(1) = -5 \\ \det A_{32} &= (2)(2) - (3)(3) = -5. \end{aligned}$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = -5$.

Exercise 3.1.4: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 1 \\ 3 & 4 & 2 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (1)(-1)^{1+1}\begin{vmatrix} 1 & 1 \\ 4 & 2 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + (5)(-1)^{1+3}\begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} = 20.$$

The 2×2 determinants used above are

$$\det A_{11} = (1)(2) - (1)(4) = -2$$
$$\det A_{12} = (2)(2) - (1)(3) = 1$$
$$\det A_{13} = (2)(4) - (1)(3) = 5.$$

Check by expanding down column 2, as requested.

$$\det A = (3)(-1)^{1+2}\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} + (1)(-1)^{2+2}\begin{vmatrix} 1 & 5 \\ 3 & 2 \end{vmatrix} + (4)(-1)^{3+2}\begin{vmatrix} 1 & 5 \\ 2 & 1 \end{vmatrix} = 20.$$

The 2×2 determinants used above are

$$\det A_{12} = (2)(2) - (1)(3) = 1$$
$$\det A_{22} = (1)(2) - (5)(3) = -13$$
$$\det A_{32} = (1)(1) - (5)(2) = -9.$$

The two expansions agree, as they must: the determinant does not depend on the chosen expansion.

Conclusion. $\det A = 20$.

Exercise 3.1.5: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 2 & 3 & -4 \\ 4 & 0 & 5 \\ 5 & 1 & 6 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (2)(-1)^{1+1}\begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 4 & 5 \\ 5 & 6 \end{vmatrix} + (-4)(-1)^{1+3}\begin{vmatrix} 4 & 0 \\ 5 & 1 \end{vmatrix} = -23.$$

The 2×2 determinants used above are

$$\det A_{11} = (0)(6) - (5)(1) = -5$$
$$\det A_{12} = (4)(6) - (5)(5) = -1$$
$$\det A_{13} = (4)(1) - (0)(5) = 4.$$

Conclusion. $\det A = -23$.

Exercise 3.1.6: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 5 & -2 & 4 \\ 0 & 3 & -5 \\ 2 & -4 & 7 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (5)(-1)^{1+1}\begin{vmatrix} 3 & -5 \\ -4 & 7 \end{vmatrix} + (-2)(-1)^{1+2}\begin{vmatrix} 0 & -5 \\ 2 & 7 \end{vmatrix} + (4)(-1)^{1+3}\begin{vmatrix} 0 & 3 \\ 2 & -4 \end{vmatrix} = 1.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (3)(7) - (-5)(-4) = 1 \\ \det A_{12} &= (0)(7) - (-5)(2) = 10 \\ \det A_{13} &= (0)(-4) - (3)(2) = -6. \end{aligned}$$

Conclusion. $\det A = 1$.

Exercise 3.1.7: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 3 & 0 \\ 6 & 5 & 2 \\ 9 & 7 & 3 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (4)(-1)^{1+1}\begin{vmatrix} 5 & 2 \\ 7 & 3 \end{vmatrix} + (3)(-1)^{1+2}\begin{vmatrix} 6 & 2 \\ 9 & 3 \end{vmatrix} = 4.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (5)(3) - (2)(7) = 1 \\ \det A_{12} &= (6)(3) - (2)(9) = 0. \end{aligned}$$

Conclusion. $\det A = 4$.

Exercise 3.1.8: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 8 & 1 & 6 \\ 4 & 0 & 3 \\ 3 & -2 & 5 \end{bmatrix}.$$

Expand across row 1.

$$\det A = (8)(-1)^{1+1}\begin{vmatrix} 0 & 3 \\ -2 & 5 \end{vmatrix} + (1)(-1)^{1+2}\begin{vmatrix} 4 & 3 \\ 3 & 5 \end{vmatrix} + (6)(-1)^{1+3}\begin{vmatrix} 4 & 0 \\ 3 & -2 \end{vmatrix} = -11.$$

The 2×2 determinants used above are

$$\det A_{11} = (0)(5) - (3)(-2) = 6$$
$$\det A_{12} = (4)(5) - (3)(3) = 11$$
$$\det A_{13} = (4)(-2) - (0)(3) = -8.$$

Conclusion. $\det A = -11$.

Exercise 3.1.9: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix} = (2)(-1)^{3+1} D_2 = 10.$$

$$D_2 = \begin{vmatrix} 0 & 0 & 5 \\ 7 & 2 & -5 \\ 3 & 1 & 8 \end{vmatrix} = (5)(-1)^{1+3} D_3 = 5.$$

$$D_3 = \begin{vmatrix} 7 & 2 \\ 3 & 1 \end{vmatrix} = (7)(1) - (2)(3) = 1.$$

Thus the answer is 10.

Exercise 3.1.10: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{vmatrix} = (3)(-1)^{2+3} D_2 = -6.$$

$$D_2 = \begin{vmatrix} 1 & -2 & 2 \\ 2 & -6 & 5 \\ 5 & 0 & 4 \end{vmatrix} = (5)(-1)^{3+1} D_3 + (4)(-1)^{3+3} D_4 = 2.$$

$$D_4 = \begin{vmatrix} 1 & -2 \\ 2 & -6 \end{vmatrix} = (1)(-6) - (-2)(2) = -2.$$

$$D_3 = \begin{vmatrix} -2 & 2 \\ -6 & 5 \end{vmatrix} = (-2)(5) - (2)(-6) = 2.$$

Thus the answer is -6.

Exercise 3.1.11: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 3 & 5 & -8 & 4 \\ 0 & -2 & 3 & -7 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 3 & 5 & -8 & 4 \\ 0 & -2 & 3 & -7 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 2 \end{vmatrix} = (2)(-1)^{4+4} D_2 = -12.$$

$$D_2 = \begin{vmatrix} 3 & 5 & -8 \\ 0 & -2 & 3 \\ 0 & 0 & 1 \end{vmatrix} = (1)(-1)^{3+3} D_3 = -6.$$

$$D_3 = \begin{vmatrix} 3 & 5 \\ 0 & -2 \end{vmatrix} = (3)(-2) - (5)(0) = -6.$$

Thus the answer is -12.

Exercise 3.1.12: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the

deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{vmatrix} = (4)(-1)^{1+1} D_2 = 36.$$

$$D_2 = \begin{vmatrix} -1 & 0 & 0 \\ 6 & 3 & 0 \\ -8 & 4 & -3 \end{vmatrix} = (-1)(-1)^{1+1} D_3 = 9.$$

$$D_3 = \begin{vmatrix} 3 & 0 \\ 4 & -3 \end{vmatrix} = (3)(-3) - (0)(4) = -9.$$

Thus the answer is 36.

Exercise 3.1.13: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 4 & 0 & -7 & 3 & -5 \\ 0 & 0 & 2 & 0 & 0 \\ 7 & 3 & -6 & 4 & -8 \\ 5 & 0 & 5 & 2 & -3 \\ 0 & 0 & 9 & -1 & 2 \end{vmatrix} = (2)(-1)^{2+3} D_2 = 6.$$

$$D_2 = \begin{vmatrix} 4 & 0 & 3 & -5 \\ 7 & 3 & 4 & -8 \\ 5 & 0 & 2 & -3 \\ 0 & 0 & -1 & 2 \end{vmatrix} = (3)(-1)^{2+2} D_3 = -3.$$

$$D_3 = \begin{vmatrix} 4 & 3 & -5 \\ 5 & 2 & -3 \\ 0 & -1 & 2 \end{vmatrix} = (-1)(-1)^{3+2} D_4 + (2)(-1)^{3+3} D_5 = -1.$$

$$D_5 = \begin{vmatrix} 4 & 3 \\ 5 & 2 \end{vmatrix} = (4)(2) - (3)(5) = -7.$$

$$D_4 = \begin{vmatrix} 4 & -5 \\ 5 & -3 \end{vmatrix} = (4)(-3) - (-5)(5) = 13.$$

Thus the answer is 6.

Exercise 3.1.14: Cofactor expansion

What we must show. Evaluate the determinant of

$$A = \begin{bmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{bmatrix}.$$

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{vmatrix} = (3)(-1)^{4+1} D_2 = 9.$$

$$D_2 = \begin{vmatrix} 3 & 2 & 4 & 0 \\ 0 & -4 & 1 & 0 \\ -5 & 6 & 7 & 1 \\ 2 & 3 & 2 & 0 \end{vmatrix} = (1)(-1)^{3+4} D_3 = -3.$$

$$D_3 = \begin{vmatrix} 3 & 2 & 4 \\ 0 & -4 & 1 \\ 2 & 3 & 2 \end{vmatrix} = (-4)(-1)^{2+2} D_4 + (1)(-1)^{2+3} D_5 = 3.$$

$$D_5 = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} = (3)(3) - (2)(2) = 5.$$

$$D_4 = \begin{vmatrix} 3 & 4 \\ 2 & 2 \end{vmatrix} = (3)(2) - (4)(2) = -2.$$

Thus the answer is 9.

Exercise 3.1.15: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 3 & 0 & 4 \\ 2 & 3 & 2 \\ 0 & 5 & -1 \end{bmatrix}$. The three downward products are -9, 0, 40, and the three upward products are 0, 30, 0. Add the downward products and subtract the entire upward sum:

$$\det A = (-9 + 0 + 40) - (0 + 30 + 0) = 1.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.16: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 0 & 5 & 1 \\ 4 & -3 & 0 \\ 2 & 4 & 1 \end{bmatrix}$. The three downward products are 0, 0, 16, and the three upward products are -6, 0, 20. Add the downward products and subtract the entire upward sum:

$$\det A = (0 + 0 + 16) - (-6 + 0 + 20) = 2.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.17: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 2 & -4 & 3 \\ 3 & 1 & 2 \\ 1 & 4 & -1 \end{bmatrix}$. The three downward products are -2, -8, 36, and the three upward products are 3, 16, 12. Add the downward products and subtract the entire upward sum:

$$\det A = (-2 + -8 + 36) - (3 + 16 + 12) = -5.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.18: The six diagonal products

What we must show. Use the six-diagonal rule on $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 1 & 1 \\ 3 & 4 & 2 \end{bmatrix}$. The three downward products are 2, 9, 40, and the three upward products are 15, 4, 12. Add the downward products and subtract the entire upward sum:

$$\det A = (2 + 9 + 40) - (15 + 4 + 12) = 20.$$

This shortcut is an arrangement of the six signed terms in a 3×3 determinant; it is not a rule for larger matrices.

Exercise 3.1.19: A row operation and its determinant

The second matrix is obtained by $R_1 \leftrightarrow R_2$. Its determinant is $cb - da = -(ad - bc)$. Thus a row interchange reverses the determinant's sign.

Exercise 3.1.20: A row operation and its determinant

The operation is $R_2 \leftarrow kR_2$. Direct calculation gives $a(kd) - b(kc) = k(ad - bc)$. Multiplying one row by k multiplies the determinant by k.

Exercise 3.1.21: A row operation and its determinant

The operation is $R_2 \leftarrow R_2 + kR_1$. The old determinant is $3 \cdot 6 - 4 \cdot 5 = -2$; the new one is $3(6 + 4k) - 4(5 + 3k) = 18 + 12k - 20 - 12k = -2$. The added terms cancel, so the determinant is unchanged.

Exercise 3.1.22: A row operation and its determinant

The operation is $R_1 \leftarrow R_1 + kR_2$. The new determinant is $(a+kc)d-(b+kd)c = ad+kcd-bc-kdc = ad - bc$. Thus row replacement preserves the determinant.

Exercise 3.1.23: A row operation and its determinant

The first row $(1, 1, 1)$ is replaced by $(k, k, k) = k(1, 1, 1)$. To see the factor directly, expand along row 1: every term in the expansion gains exactly one factor k. Thus the new determinant is k times the old one (also when $k = 0$).

Exercise 3.1.24: A row operation and its determinant

The operation interchanges rows 1 and 2. Interchanging these two rows reverses the sign of every signed product in the determinant, so the new determinant is the negative of the old determinant.

Exercise 3.1.25: An elementary-matrix determinant

This lower triangular matrix has diagonal $1, 1, 1$. Its determinant is their product, 1. It performs $R_3 \leftarrow R_3 + kR_2$ on another matrix.

Exercise 3.1.26: An elementary-matrix determinant

The matrix is lower triangular with diagonal $1, 1, 1$, so its determinant is 1. Its row operation is $R_3 \leftarrow R_3 + kR_1$.

Exercise 3.1.27: An elementary-matrix determinant

The diagonal matrix has diagonal $k, 1, 1$, so its determinant is k. It scales row 1 by k. For an elementary matrix this operation requires $k \neq 0$; the displayed determinant formula is still valid at $k = 0$.

Exercise 3.1.28: An elementary-matrix determinant

The diagonal entries are $1, k, 1$, giving determinant k. The operation scales row 2 by k; an elementary scaling has $k \neq 0$.

Exercise 3.1.29: An elementary-matrix determinant

The matrix is obtained from I_3 by interchanging rows 1 and 2, so its determinant is -1. Direct expansion along row 3 gives $1 \det \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right] = -1$.

Exercise 3.1.30: An elementary-matrix determinant

The matrix interchanges rows 1 and 3 of I_3. Its determinant is -1. Expansion across row 2 gives $1(-1)^{2+2} \det \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right] = -1$.

Exercise 3.1.31: An elementary-matrix determinant

A row-replacement elementary matrix is obtained from I_n by inserting a scalar k into an off-diagonal position (i, j). Expand by rows different from i, j to reduce to the 2×2 block $\left[\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right]$ or its transpose. The determinant is 1. Equivalently, adding k times one identity row to another creates no new nonzero signed product. The examples in Exercises 25 and 26 illustrate this general result.

Exercise 3.1.32: An elementary-matrix determinant

A scaling elementary matrix is diagonal with one diagonal entry $k \neq 0$ and all others 1. The triangular-matrix theorem therefore gives determinant k. It is one row that is scaled, not all n rows.

Exercise 3.1.33: Verify multiplication for one elementary matrix

Here $E = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, so $EA = \left[\begin{smallmatrix} c & d \\ a & b \end{smallmatrix}\right]$. Consequently $\det(EA) = cb - da = -(ad - bc)$, while $\det E = -1$. Thus $\det(EA) = (\det E)(\det A)$.

Exercise 3.1.34: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & 0 \\ 0 & k \end{smallmatrix}\right]$, we have $EA = \left[\begin{smallmatrix} a & b \\ kc & kd \end{smallmatrix}\right]$. Its determinant is $akd - bkc = k(ad - bc)$. Since $\det E = k$, the required equality follows.

Exercise 3.1.35: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right]$, multiplication gives $EA = \left[\begin{smallmatrix} a+kc & b+kd \\ c & d \end{smallmatrix}\right]$. Its determinant is $ad+kcd-bc-kdc = ad-bc$. Also $\det E = 1$, so the two sides agree.

Exercise 3.1.36: Verify multiplication for one elementary matrix

For $E = \left[\begin{smallmatrix} 1 & 0 \\ k & 1 \end{smallmatrix}\right]$, multiplication gives $EA = \left[\begin{smallmatrix} a & b \\ ka+c & kb+d \end{smallmatrix}\right]$. Its determinant is $a(kb+d)-b(ka+c) = ad-bc$. This equals $(\det E)(\det A)$ because $\det E = 1$.

Exercise 3.1.37: Determinants and their interpretation

Scalar multiplication scales every entry, so $5A = \left[\begin{smallmatrix} 15 & 5 \\ 20 & 10 \end{smallmatrix}\right]$. We find $\det A = 3 \cdot 2 - 1 \cdot 4 = 2$, but $\det(5A) = 15 \cdot 10 - 5 \cdot 20 = 50$, whereas $5 \det A = 10$. Thus the proposed equality is false. Both rows were multiplied by 5, giving a factor 5^2.

Exercise 3.1.38: Determinants and their interpretation

Write $kA = \left[\begin{smallmatrix} ka & kb \\ kc & kd \end{smallmatrix}\right]$. Then $\det(kA) = (ka)(kd) - (kb)(kc) = k^2(ad - bc) = k^2 \det A$. This calculation also covers $k = 0$.

Exercise 3.1.39: Determinants and their interpretation

(a) True, with the usual initial case.

For $n \geq 2$, the recursive definition expands into determinants of $(n-1) \times (n-1)$ minor matrices. A 1×1 determinant is its sole entry, the base case.

(b) False: a cofactor is a scalar.

Deleting a row and column produces the matrix A_{ij}. The cofactor is $(-1)^{i+j} \det A_{ij}$, a signed number, not that submatrix.

Exercise 3.1.40: Determinants and their interpretation

(a) False.

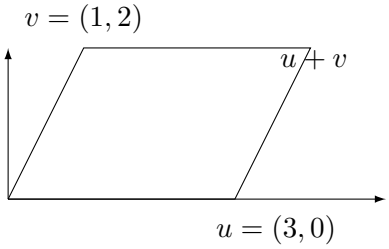
Row and column expansions compute the same number $\det A$, not negatives in general. For example, both expansions of I_2 give 1.

(b) False.

For a triangular matrix the determinant is the *product* of diagonal entries. For I_2 the determinant is 1, whereas the sum of the diagonal is 2.

Exercise 3.1.41: Determinants and their interpretation

What we must show. Compare unsigned area with a determinant, and then vary the shear. The two adjacent edges are $u = (3, 0)$ and $v = (1, 2)$. The horizontal base has length 3 and perpendicular height 2, so the area is $3 \cdot 2 = 6$. Also $\det[u\ v] = 3 \cdot 2 - 0 \cdot 1 = 6$. Replacing v by $(x, 2)$ leaves its height above the horizontal base equal to 2, hence leaves the area 6; the determinant is still $3 \cdot 2 - 0x = 6$.

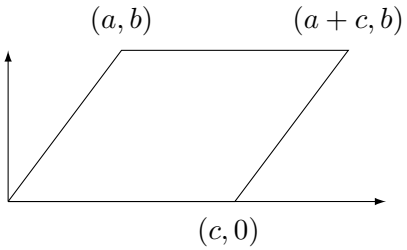


The horizontal displacement changes the slant, not the height. This operation is a shear; the sketch uses $x = 1$, and the argument works for every real x.

Exercise 3.1.42: Determinants and their interpretation

Take the horizontal base $v = (c, 0)$ of length $c > 0$. The other edge $u = (a, b)$ has height $b > 0$, so the area is bc. On the other hand,

$$\det[u\ v] = \begin{vmatrix} a & c \\ b & 0 \end{vmatrix} = -bc, \qquad \det[v\ u] = \begin{vmatrix} c & a \\ 0 & b \end{vmatrix} = bc.$$



The sketch is schematic. Interchanging the order of the columns changes orientation and the determinant's sign, but the ordinary area is nonnegative and equals either determinant's absolute value.

Exercise 3.1.43: A reproducible test of determinant additivity

What we must show. Test additivity with four reproducible randomly generated pairs. A computer experiment supplies evidence, not a proof of a universal identity. Here is an exact integer experiment; a fixed seed lets another reader repeat the same draws.

```
from random import Random
from sympy import Matrix
r = Random(31043)
for n in (5,2,3,4):
    A = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    B = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    print(n, A.det(), B.det(), (A+B).det(),
          (A+B).det()-A.det()-B.det())
```

n	$\det A$	$\det B$	$\det(A+B)$	difference
5	10070	136248	121574	−24744
2	−34	1	−78	−45
3	888	−55	1464	631
4	176	748	−6819	−7743

The nonzero differences disprove additivity. A particularly short exact counterexample is $A = B = I_2$: $\det(A + B) = 4$, but $\det A + \det B = 2$. The determinant is linear in *one row at a time with the other rows fixed*, not linear in the entire matrix.

Exercise 3.1.44: A reproducible test of multiplication

Use the same four random pairs and seed from Exercise 43. Replace the last line of that code by

```
print(n, (A*B).det(), A.det()*B.det(),
      (A*B).det()-A.det()*B.det())
```

The exact outputs are

n	$\det(AB)$	$(\det A)(\det B)$	difference
5	1372017360	1372017360	0
2	−34	−34	0
3	−48840	−48840	0
4	131648	131648	0

All four differences are zero. This suggests $\det(AB) = (\det A)(\det B)$. The next section proves it for all square matrices, so the experiment is not being substituted for a proof.

Exercise 3.1.45: Scaling and transposition in several dimensions

Generate three 4×4 samples and two each of sizes 5 and 6. The following code carries out all comparisons, including transpose, with exact arithmetic.

```
from random import Random
from sympy import Matrix
r = Random(31045)
for n in (4,4,4,5,5,6,6):
    A = Matrix(n,n,[r.randint(-9,9) for _ in range(n*n)])
    d = A.det()
    assert A.T.det() == d
    print(n,d,(-A).det(),(2*A).det(),(10*A).det())
```

The transpose check passed for every sample; the remaining outputs are

n	$\det A$	$\det(-A)$	$\det(2A)$	$\det(10A)$
4	2446	2446	39136	24460000
4	3572	3572	57152	35720000
4	2144	2144	34304	21440000
5	150900	−150900	4828800	15090000000
5	29441	−29441	942112	2944100000
6	−122640	−122640	−7848960	−122640000000
6	−191568	−191568	−12260352	−191568000000

The conjectures are $\det A^T = \det A$ and $\det(kA) = k^n \det A$. In particular, negation preserves the determinant in even dimensions and reverses its sign in odd dimensions. Scaling the matrix scales *every* row, so the scale factor occurs n times. The integer calculations above check that description at every sampled size.

Exercise 3.1.46: A determinant of an inverse

Use seed 31046 and the same integer generator as Exercise 45, now at sizes $4, 5, 6$. For each generated matrix, compute

```
d = A.det()
if d != 0:
    print(n,d,A.inv().det())
else:
    print(A.echelon_form())
```

The three sampled matrices were nonsingular and gave

n	$\det A$	$\det(A^{-1})$
4	-1586	$-\frac{1}{1586}$
5	34641	$\frac{1}{34641}$
6	-48084	$-\frac{1}{48084}$

Each last entry is the reciprocal of the preceding one. Thus the conjecture is $\det(A^{-1}) = 1/\det A$ when A is invertible. A zero determinant would instead lead to a missing pivot in echelon form and no inverse; the reciprocal expression would then be undefined. Multiplicativity in the next section proves the conjecture by applying it to $AA^{-1} = I$.

3.2 Properties of determinants

A row replacement preserves a determinant, a row interchange reverses its sign, and multiplying a single row by k multiplies it by k. In the calculations below, keeping the operations visible is how we avoid losing a sign or a scale factor. Once a matrix is triangular its determinant is the product of its diagonal entries.

A square matrix is invertible exactly when its determinant is nonzero. Thus n vectors in $\mathbb{R}^n$ are linearly independent exactly when the determinant of the matrix having those vectors as columns is nonzero. The multiplication theorem says $\det(AB) = \det A \det B$, and transposition preserves a determinant.

Practice problem 3.2.P1: Create a zero row

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_4 \leftarrow R_4 + 3R_1$. Both resulting rows 2 and 4 are $(0, 1, -3, 2)$. Then $R_4 \leftarrow R_4 - R_2$ creates a zero row. All three operations preserve the determinant, and expansion along the zero row gives $\det A = 0$.

Practice problem 3.2.P2: A determinant tests independence

Place the three vectors in columns, not rows:

$$A = \begin{bmatrix} 5 & -3 & 2 \\ -7 & 3 & -7 \\ 9 & -5 & 5 \end{bmatrix}.$$

Replace R_2 by $R_2 + R_1 = (-2, 0, -5)$. Expanding down column 2 now gives

$$\det A = 3\begin{vmatrix} -2 & -5 \\ 9 & 5 \end{vmatrix} + 5\begin{vmatrix} 5 & 2 \\ -2 & -5 \end{vmatrix} = 3(35) + 5(-21) = 0.$$

Thus A is not invertible, and the three columns are linearly dependent by the Invertible Matrix Theorem.

Exercise 3.2.1: Identify the determinant rule

The displayed matrices differ by $R_1 \leftrightarrow R_2$. A single interchange reverses the determinant's sign, which explains the minus sign in the equation.

Exercise 3.2.2: Identify the determinant rule

The first row $(2, -6, 4)$ is $2(1, -3, 2)$, while the other rows are fixed. Linearity in one row gives exactly the displayed factor 2.

Exercise 3.2.3: Identify the determinant rule

The right-hand matrix results from $R_2 \leftarrow R_2 - 2R_1$, since $(2, 0, -3) - 2(1, 3, -4) = (0, -6, 5)$. A row replacement preserves the determinant.

Exercise 3.2.4: Identify the determinant rule

Apply $R_3 \leftarrow R_3 - 3R_1$: $(3,7,4) - 3(1,2,3) = (0,1,-5)$. This is a row replacement and does not change the determinant.

Exercise 3.2.5: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & -6 \\ -1 & -4 & 4 \\ -2 & -7 & 9 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-1)R_1, \quad R_3 \leftarrow R_3 - (-2)R_1$, giving

$$\begin{bmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 3 & -3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (3)R_2$, giving

$$\begin{bmatrix} 1 & 5 & -6 \\ 0 & 1 & -2 \\ 0 & 0 & 3 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(3) = 3.$$

Exercise 3.2.6: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 5 & -3 \\ 3 & -3 & 3 \\ 2 & 13 & -7 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (3)R_1, \quad R_3 \leftarrow R_3 - (2)R_1$, giving

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & -18 & 12 \\ 0 & 3 & -1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{1}{6})R_2$, giving

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & -18 & 12 \\ 0 & 0 & 1 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(-18)(1) = -18.$$

Exercise 3.2.7: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-2)R_1, \quad R_3 \leftarrow R_3 - (3)R_1, \quad R_4 \leftarrow R_4 - (1)R_1$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & -4 & 2 & -5 \\ 0 & -4 & 2 & -5 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-4)R_2, \quad R_4 \leftarrow R_4 - (-4)R_2$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & 0 & 30 & 27 \\ 0 & 0 & 30 & 27 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (1)R_3$, giving

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7 & 8 \\ 0 & 0 & 30 & 27 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(30)(0) = 0.$$

Exercise 3.2.8: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 2 & 5 & 4 & -3 \\ -3 & -7 & -5 & 2 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3 - (2)R_1, \quad R_4 \leftarrow R_4 - (-3)R_1$, giving

$$\begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 0 & -1 & -2 & 5 \\ 0 & 2 & 4 & -10 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-1)R_2, \quad R_4 \leftarrow R_4 - (2)R_2$, giving

$$\begin{bmatrix} 1 & 3 & 3 & -4 \\ 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(1)(1)(0)(0) = 0.$$

Exercise 3.2.9: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ -1 & 2 & 8 & 5 \\ 3 & -1 & -2 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3 - (-1)R_1, \quad R_4 \leftarrow R_4 - (3)R_1$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 1 & 5 & 5 \\ 0 & 2 & 7 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (1)R_2, \quad R_4 \leftarrow R_4 - (2)R_2$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -3 & -5 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 1 & -1 & -3 & 0 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & -3 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(1)(1)(-3)(1) = 3.$$

Exercise 3.2.10: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ -2 & -6 & 2 & 3 & 9 \\ 3 & 7 & -3 & 8 & -7 \\ 3 & 5 & 5 & 2 & 7 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_3 \leftarrow R_3-(-2)R_1, \quad R_4 \leftarrow R_4-(3)R_1, \quad R_5 \leftarrow R_5-(3)R_1$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & -2 & 0 & 8 & -1 \\ 0 & -4 & 8 & 2 & 13 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4-(-1)R_2, \quad R_5 \leftarrow R_5-(-2)R_2$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & 0 & -4 & 7 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 1 & 3 & -1 & 0 & -2 \\ 0 & 2 & -4 & -1 & -6 \\ 0 & 0 & -4 & 7 & -7 \\ 0 & 0 & 0 & 3 & 5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(1)(2)(-4)(3)(1) = 24.$$

Exercise 3.2.11: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2-(\frac{3}{2})R_1, \quad R_3 \leftarrow R_3-(-3)R_1, \quad R_4 \leftarrow R_4-(2)R_1$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 15 & -13 & 6 \\ 0 & 0 & 2 & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3-(-2)R_2$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 2 & 1 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-1)R_3$, giving

$$\begin{bmatrix} 2 & 5 & -3 & -1 \\ 0 & -\frac{15}{2} & \frac{11}{2} & -\frac{3}{2} \\ 0 & 0 & -2 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-\frac{15}{2})(-2)(4) = 120.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.12: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ 5 & 4 & 6 & 6 \\ 4 & 2 & 4 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-3)R_1, \quad R_3 \leftarrow R_3 - (-5)R_1, \quad R_4 \leftarrow R_4 - (-4)R_1$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 14 & 21 & 6 \\ 0 & 10 & 16 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{7}{5})R_2, \quad R_4 \leftarrow R_4 - (1)R_2$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 0 & \frac{21}{5} & 6 \\ 0 & 0 & 4 & 3 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (\frac{20}{21})R_3$, giving

$$\begin{bmatrix} -1 & 2 & 3 & 0 \\ 0 & 10 & 12 & 0 \\ 0 & 0 & \frac{21}{5} & 6 \\ 0 & 0 & 0 & -\frac{19}{7} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(-1)(10)(\frac{21}{5})(-\frac{19}{7}) = 114.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.13: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} 2 & 5 & 4 & 1 \\ 4 & 7 & 6 & 2 \\ 6 & -2 & -4 & 0 \\ -6 & 7 & 7 & 0 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (2)R_1, \quad R_3 \leftarrow R_3 - (3)R_1, \quad R_4 \leftarrow R_4 - (-3)R_1$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & -17 & -16 & -3 \\ 0 & 22 & 19 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{17}{3})R_2, \quad R_4 \leftarrow R_4 - (-\frac{22}{3})R_2$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & 0 & -\frac{14}{3} & -3 \\ 0 & 0 & \frac{13}{3} & 3 \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-\frac{13}{14})R_3$, giving

$$\begin{bmatrix} 2 & 5 & 4 & 1 \\ 0 & -3 & -2 & 0 \\ 0 & 0 & -\frac{14}{3} & -3 \\ 0 & 0 & 0 & \frac{3}{14} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-3)(-\frac{14}{3})(\frac{3}{14}) = 6.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.14: Row reduction with determinant bookkeeping

What we must show. Compute the determinant of

$$A = \begin{bmatrix} -3 & -2 & 1 & -4 \\ 1 & 3 & 0 & -3 \\ -3 & 4 & -2 & 8 \\ 3 & -4 & 0 & 4 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-\frac{1}{3})R_1, \quad R_3 \leftarrow R_3 - (1)R_1, \quad R_4 \leftarrow R_4 - (-1)R_1$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 6 & -3 & 12 \\ 0 & -6 & 1 & 0 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{18}{7})R_2, \quad R_4 \leftarrow R_4 - (-\frac{18}{7})R_2$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 0 & -\frac{27}{7} & \frac{162}{7} \\ 0 & 0 & \frac{13}{7} & -\frac{78}{7} \end{bmatrix}.$$

Perform $R_4 \leftarrow R_4 - (-\frac{13}{27})R_3$, giving

$$\begin{bmatrix} -3 & -2 & 1 & -4 \\ 0 & \frac{7}{3} & \frac{1}{3} & -\frac{13}{3} \\ 0 & 0 & -\frac{27}{7} & \frac{162}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(-3)(\frac{7}{3})(-\frac{27}{7})(0) = 0.$$

This last product can also be read as repeated cofactor expansion down the first column of the triangular matrix: each expansion removes its first row and column and leaves the next diagonal entry as a factor. Thus the calculation combines row reduction with sparse cofactor expansion.

Exercise 3.2.15: Transform a known determinant

The third row is scaled by 5 and the other rows are fixed. Thus the determinant is $5 \cdot 7 = 35$.

Exercise 3.2.16: Transform a known determinant

The second row is scaled by 3, so the determinant is $3 \cdot 7 = 21$.

Exercise 3.2.17: Transform a known determinant

Interchanging rows 2 and 3 once gives the displayed matrix. Its determinant is -7.

Exercise 3.2.18: Transform a known determinant

Move the original third row to the top by two interchanges: first interchange rows 2 and 3, then rows 1 and 2. The two sign changes cancel, leaving determinant 7.

Exercise 3.2.19: Transform a known determinant

First scale the second row by 2, multiplying the determinant by 2. Then add row 1 to it, which preserves that new determinant. The answer is $2 \cdot 7 = 14$.

Exercise 3.2.20: Transform a known determinant

The operation $R_1 \leftarrow R_1 + R_2$ preserves the determinant, so the answer is 7.

Exercise 3.2.21: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 2 & 3 & 0 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{2})R_1, \quad R_3 \leftarrow R_3 - (\frac{1}{2})R_1$, giving

$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & \frac{3}{2} & 4 \\ 0 & \frac{1}{2} & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{1}{3})R_2$, giving

$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & \frac{3}{2} & 4 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(\frac{3}{2})(-\frac{1}{3}) = -1.$$

Since the determinant is -1, the matrix is invertible by the determinant criterion.

Exercise 3.2.22: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 5 & 0 & -1 \\ 1 & -3 & -2 \\ 0 & 5 & 3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{5})R_1$, giving

$$\begin{bmatrix} 5 & 0 & -1 \\ 0 & -3 & -\frac{9}{5} \\ 0 & 5 & 3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{5}{3})R_2$, giving

$$\begin{bmatrix} 5 & 0 & -1 \\ 0 & -3 & -\frac{9}{5} \\ 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(5)(-3)(0) = 0.$$

Since the determinant is 0, the matrix is not invertible by the determinant criterion.

Exercise 3.2.23: A square-matrix determinant test

What we must show. Test invertibility of the displayed square matrix.

$$A = \begin{bmatrix} 2 & 0 & 0 & 8 \\ 1 & -7 & -5 & 0 \\ 3 & 8 & 6 & 0 \\ 0 & 7 & 5 & 4 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{1}{2})R_1, \quad R_3 \leftarrow R_3 - (\frac{3}{2})R_1$, giving

$$\begin{bmatrix} 2 & 0 & 0 & 8 \\ 0 & -7 & -5 & -4 \\ 0 & 8 & 6 & -12 \\ 0 & 7 & 5 & 4 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{8}{7})R_2, \quad R_4 \leftarrow R_4 - (-1)R_2$, giving

$$\begin{bmatrix} 2 & 0 & 0 & 8 \\ 0 & -7 & -5 & -4 \\ 0 & 0 & \frac{2}{7} & -\frac{116}{7} \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(2)(-7)(\frac{2}{7})(0) = 0.$$

Since the determinant is 0, the matrix is not invertible by the determinant criterion.

Exercise 3.2.24: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 4 & -7 & -3 \\ 6 & 0 & -5 \\ -7 & 2 & 6 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{3}{2})R_1, \quad R_3 \leftarrow R_3 - (-\frac{7}{4})R_1$, giving

$$\begin{bmatrix} 4 & -7 & -3 \\ 0 & \frac{21}{2} & -\frac{1}{2} \\ 0 & -\frac{41}{4} & \frac{3}{4} \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{41}{42})R_2$, giving

$$\begin{bmatrix} 4 & -7 & -3 \\ 0 & \frac{21}{2} & -\frac{1}{2} \\ 0 & 0 & \frac{11}{42} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(4)(\frac{21}{2})(\frac{11}{42}) = 11.$$

There are 3 vectors in $\mathbb{R}^3$, so the determinant test applies. The determinant is 11; hence the set is linearly independent.

Exercise 3.2.25: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 7 & -8 & 7 \\ -4 & 5 & 0 \\ -6 & 7 & -5 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (-\frac{4}{7})R_1, \quad R_3 \leftarrow R_3 - (-\frac{6}{7})R_1$, giving

$$\begin{bmatrix} 7 & -8 & 7 \\ 0 & \frac{3}{7} & 4 \\ 0 & \frac{1}{7} & 1 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (\frac{1}{3})R_2$, giving

$$\begin{bmatrix} 7 & -8 & 7 \\ 0 & \frac{3}{7} & 4 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix}.$$

There were 0 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^0(7)(\frac{3}{7})(-\frac{1}{3}) = -1.$$

There are 3 vectors in $\mathbb{R}^3$, so the determinant test applies. The determinant is -1; hence the set is linearly independent.

Exercise 3.2.26: A square-matrix determinant test

What we must show. Test independence by placing the given vectors in columns.

$$A = \begin{bmatrix} 3 & 2 & -2 & 0 \\ 5 & -6 & -1 & 0 \\ -6 & 0 & 3 & 0 \\ 4 & 7 & 0 & -3 \end{bmatrix}.$$

Use only row replacements and, when needed, row interchanges. A replacement preserves the determinant; an interchange multiplies it by -1. No row is scaled. Perform $R_2 \leftarrow R_2 - (\frac{5}{3})R_1, \quad R_3 \leftarrow R_3 - (-2)R_1, \quad R_4 \leftarrow R_4 - (\frac{4}{3})R_1$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 4 & -1 & 0 \\ 0 & \frac{13}{3} & \frac{8}{3} & -3 \end{bmatrix}.$$

Perform $R_3 \leftarrow R_3 - (-\frac{3}{7})R_2, \quad R_4 \leftarrow R_4 - (-\frac{13}{28})R_2$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{15}{4} & -3 \end{bmatrix}.$$

Perform $R_3 \leftrightarrow R_4$, giving

$$\begin{bmatrix} 3 & 2 & -2 & 0 \\ 0 & -\frac{28}{3} & \frac{7}{3} & 0 \\ 0 & 0 & \frac{15}{4} & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

There were 1 row interchanges. The final matrix is triangular, so

$$\det A = (-1)^1(3)(-\frac{28}{3})(\frac{15}{4})(0) = 0.$$

There are 4 vectors in $\mathbb{R}^4$, so the determinant test applies. The determinant is 0; hence the set is linearly dependent.

Exercise 3.2.27: Properties, proofs, and exact calculations

(a) True.

This is the row-replacement part of Theorem 3.

(b) False as stated for an arbitrary echelon form.

The proposed sign correction accounts for row interchanges, but not row scalings. Starting with $A = I_2$, scale row 1 by 2. The resulting echelon form has pivot product 2 and no interchanges, although $\det A = 1$. If reduction uses *only* replacements and interchanges, the stated product with its sign correction is valid.

(c) True.

Dependent columns mean that A is not invertible, by the Invertible Matrix Theorem; Theorem 4 then gives $\det A = 0$.

(d) False.

Take $A = B = I_2$. The left side is $\det(2I_2) = 4$, while the right side is $1 + 1 = 2$.

Exercise 3.2.28: Properties, proofs, and exact calculations

(a) True.

Each interchange multiplies the determinant by -1, so two multiply it by $(-1)^2 = 1$.

(b) False.

The diagonal-product rule needs a triangular matrix. For example, $\left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ has determinant 0, but diagonal product 1.

(c) False.

The matrix $\left[\begin{smallmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{smallmatrix}\right]$ has row 3 equal to row 1 plus row 2 and therefore determinant zero. None of its rows or columns is zero or identical to another.

(d) False.

Theorem 5 gives $\det A^T = \det A$. The identity matrix, whose determinant is 1, disproves the proposed minus sign.

Exercise 3.2.29: Properties, proofs, and exact calculations

For $B = \left[\begin{smallmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{smallmatrix}\right]$, subtract row 1 from rows 2 and 3, giving lower right block $\left[\begin{smallmatrix} 1 & 1 \\ 2 & 0 \end{smallmatrix}\right]$. Thus $\det B = 1(0 - 2) = -2$. Applying multiplicativity repeatedly gives $\det(B^5) = (\det B)^5 = (-2)^5 = -32$.

Exercise 3.2.30: Properties, proofs, and exact calculations

Interchange the two equal rows. The numerical matrix does not change, but Theorem 3 says its determinant changes sign. Thus $d = -d$, where $d = \det A$. Adding d gives $2d = 0$, hence $d = 0$. If two columns are equal, transpose the matrix; its transpose has two equal rows and the same determinant by Theorem 5. This proof does not use the invertibility criterion in Theorem 4.

Exercise 3.2.31: Properties, proofs, and exact calculations

Since A is invertible, Theorem 4 gives $\det A \neq 0$. The multiplication theorem applied to $AA^{-1} = I$ gives $(\det A)(\det A^{-1}) = \det I = 1$. Dividing by the nonzero scalar $\det A$ proves $\det A^{-1} = 1/\det A$.

Exercise 3.2.32: Properties, proofs, and exact calculations

The matrix rA is obtained by multiplying each of the n rows by r. Repeated use of Theorem 3(c), one row at a time, gives $\det(rA) = r^n \det A$. If $r = 0$ and $n \geq 1$, the matrix is zero and both sides are zero, so that case is also covered.

Exercise 3.2.33: Properties, proofs, and exact calculations

Theorem 6 and commutativity of *real-number* multiplication give $\det(AB) = (\det A)(\det B) = (\det B)(\det A) = \det(BA)$. This does not assert that the matrix products themselves commute.

Exercise 3.2.34: Properties, proofs, and exact calculations

By multiplicativity, $\det(PAP^{-1}) = (\det P)(\det A)(\det P^{-1})$. Because P is invertible, Exercise 31 gives $(\det P)(\det P^{-1}) = 1$. The product therefore reduces to $\det A$. Similar matrices have the same determinant.

Exercise 3.2.35: Properties, proofs, and exact calculations

Take determinants of $U^TU = I$. Theorems 5 and 6 give $(\det U)^2 = (\det U^T)(\det U) = 1$. Since $\det U$ is real, solving $d^2 = 1$ gives $d = 1$ or $d = -1$. Both signs occur: I gives 1, and a single coordinate reflection gives -1.

Exercise 3.2.36: Properties, proofs, and exact calculations

Repeated multiplication gives $0 = \det(A^4) = (\det A)^4$. A real number has fourth power zero only when it is zero. Thus $\det A = 0$, so Theorem 4 rules out invertibility.

Exercise 3.2.37: Properties, proofs, and exact calculations

Multiply the matrices first:

$$AB = \begin{bmatrix} 6 & 0 \\ 17 & 4 \end{bmatrix}.$$

Direct 2×2 calculations give $\det A = 3$, $\det B = 8$, and

$$\det(AB) = (6)(4) - (0)(17) = 24 = (\det A)(\det B).$$

This is a direct verification for these matrices, rather than an application of Theorem 6.

Exercise 3.2.38: Properties, proofs, and exact calculations

Multiply the matrices first:

$$AB = \begin{bmatrix} 6 & 0 \\ -2 & 0 \end{bmatrix}.$$

Direct 2×2 calculations give $\det A = 0$, $\det B = -2$, and

$$\det(AB) = (6)(0) - (0)(-2) = 0 = (\det A)(\det B).$$

This is a direct verification for these matrices, rather than an application of Theorem 6.

Exercise 3.2.39: Properties, proofs, and exact calculations

Use the given 3×3 dimension when scaling every row:

(a) $\det(AB) = 4(-3) = -12$, by multiplicativity.

(b) $\det(5A) = 5^3 \cdot 4 = 500$, since all three rows are scaled.

(c) $\det(B^T) = \det B = -3$, by transposition invariance.

(d) $\det(A^{-1}) = 1/4$, since $\det A \neq 0$.

(e) $\det(A^3) = 4^3 = 64$, by applying multiplicativity twice.

Exercise 3.2.40: Properties, proofs, and exact calculations

The matrices are now 4×4:

(a) $\det(AB) = (-1)(2) = -2$.

(b) $\det(B^5) = 2^5 = 32$.

(c) $\det(2A) = 2^4(-1) = -16$.

(d) $\det(A^T A) = \det(A^T) \det A = (-1)^2 = 1$.

(e) $\det(B^{-1}AB) = (1/2)(-1)(2) = -1$; the inverse exists because $\det B = 2 \neq 0$.

Exercise 3.2.41: Properties, proofs, and exact calculations

Expand each 2×2 determinant directly:

$$\det A = (a + e)d - (b + f)c = (ad - bc) + (ed - fc) = \det B + \det C.$$

Only the *first row* was split, while the second row stayed fixed. This is compatible with failure of additivity in the whole matrix.

Exercise 3.2.42: Properties, proofs, and exact calculations

Here $A = I_2$. Direct calculation yields

$$\det(I_2 + B) = (1 + a)(1 + d) - bc = 1 + ad - bc + (a + d).$$

But $\det I_2 + \det B = 1 + ad - bc$. The two expressions are equal if and only if their difference $a + d$ is zero. This proves both implications of the claimed equivalence.

Exercise 3.2.43: Properties, proofs, and exact calculations

Expand each determinant down its third column. The cofactors C_{13}, C_{23}, C_{33} depend only on the first two columns, so they are the same for all three matrices. Therefore

$$\begin{aligned}\det A &= \sum_{i=1}^{3}(u_i + v_i)C_{i3} \\ &= \sum_{i=1}^{3} u_i C_{i3} + \sum_{i=1}^{3} v_i C_{i3} = \det B + \det C.\end{aligned}$$

The matrices do not satisfy $A = B + C$, because adding B and C would double their first two columns. The equality is linearity in one column with the other columns held fixed.

Exercise 3.2.44: Properties, proofs, and exact calculations

Transpose first: $(AE)^T = E^T A^T$. The transpose E^T is an elementary matrix of the corresponding type. Theorem 3, applied to left multiplication by E^T, gives

$$\det(E^T A^T) = (\det E^T)(\det A^T).$$

By Theorem 5, the left side equals $\det(AE)$, and the right side equals $(\det E)(\det A)$. This argument uses row-operation rules and transpose invariance, not the general product theorem.

Exercise 3.2.45: Rectangular matrices and two Gram products

Run the following exact experiment, including two samples at each requested size.

```
from random import Random
from sympy import Matrix
r = Random(32045)
for m,n in ((4,5),(4,5),(5,6),(5,6)):
    A = Matrix(m,n,[r.randint(-5,5) for _ in range(m*n)])
    print(m,n,(A.T*A).det(),(A*A.T).det())
```

The computed outputs are

shape of A	$\det(A^T A)$	$\det(AA^T)$
4×5	0	22645
4×5	0	128772
5×6	0	33391308
5×6	0	1276024

Explain what must hold for every such matrix.

If A has $m < n$, its n columns in $\mathbb{R}^m$ are dependent, so some nonzero x satisfies $Ax = 0$. Then $A^T Ax = 0$, proving that $A^T A$ is singular and its determinant is zero. In contrast, AA^T is invertible exactly when the rows of A are independent: for any y,

$$y^T AA^T y = \|A^T y\|^2.$$

This is zero exactly when $A^T y = 0$. Thus independent rows imply a zero null space for AA^T. A dependent-row sample would also give $\det(AA^T) = 0$; its nonzero values in these four experiments are not a universal guarantee.

Exercise 3.2.46: Small determinant versus near singularity

What we must show. Compare scaling-sensitive determinants with a scaling-invariant condition number. The specified matrix from Section 2.3, Exercise 9 is

$$A = \begin{bmatrix} 4 & 0 & -3 & -7 \\ -6 & 9 & 9 & 9 \\ 7 & -5 & 10 & 19 \\ -1 & 2 & 4 & -1 \end{bmatrix}.$$

Using the spectral (2-norm) condition number, the following double-precision calculation gives the table below.

```
import numpy as np
A = np.array([[4,0,-3,-7],[-6,9,9,9],
              [7,-5,10,19],[-1,2,4,-1]],float)
for B in (A,np.eye(4)):
    for c in (0.1,1.,10.):
        print(c,np.linalg.det(c*B),np.linalg.cond(c*B,2))
```

The determinants in the table were additionally checked in exact rational arithmetic, avoiding cancellation errors in their reported values.

matrix	det	κ_2
$\frac{1}{10}A$	$-\frac{501}{1250}$	11.3525
$1A$	-4008	11.3525
$10A$	-40080000	11.3525
$\frac{1}{10}I_4$	$\frac{1}{10000}$	1
$1I_4$	1	1
$10I_4$	10000	1

Interpret the comparison.

The supplied A is less well-conditioned than the identity, with a moderate condition number about 11.35. The exercise calls it nearly singular, but the actual supplied entries do not exhibit severe ill-conditioning. Its determinant changes by a factor 10^4 when A is multiplied by 10, while its condition number does not change. In general, $\det(cA) = c^4 \det A$ in this dimension. But for nonzero c,

$$\kappa_2(cA) = \|cA\|_2 \|(cA)^{-1}\|_2 = |c| \|A\|_2 \, |c|^{-1} \|A^{-1}\|_2 = \kappa_2(A).$$

Every nonzero scalar multiple of the identity has condition number 1, even if its determinant is made arbitrarily small. Thus determinant size alone, without controlling matrix scale, does not measure relative nearness to singularity. Exact zero still characterizes singularity; "small" is the scale-dependent part.

3.3 Cramer's rule, volume, and linear transformations

Cramer's rule solves $Ax = b$ when the square coefficient matrix satisfies $\det A \neq 0$: replace column i of A by b to form $A_i(b)$, then $x_i = \det A_i(b)/\det A$. Replacing a *row* is not Cramer's rule. The adjugate is the transpose of the cofactor matrix, and $A^{-1} = \operatorname{adj}(A)/\det A$ for an invertible A.

In geometry, determinants carry orientation signs. Ordinary areas and volumes do not: a parallelogram's area is $|\det[u\ v]|$, a parallelepiped's volume is $|\det[u\ v\ w]|$, and a linear transformation multiplies area or volume by the absolute value of its determinant. Translations do not change these measures.

Practice problem 3.3.P1: Transform a parallelogram

The edge matrix is $B = \left[\begin{smallmatrix} 1 & 5 \\ 3 & 1 \end{smallmatrix}\right]$, whose determinant is $1 - 15 = -14$. Thus the original area is 14. The transformation matrix $A = \left[\begin{smallmatrix} 1 & -0.1 \\ 0 & 2 \end{smallmatrix}\right]$ has determinant 2. Consequently the image area is $|\det A|\,|\det B| = 2 \cdot 14 = 28$. As a check, $AB = \left[\begin{smallmatrix} 0.7 & 4.9 \\ 6 & 2 \end{smallmatrix}\right]$ has determinant $1.4 - 29.4 = -28$, whose absolute value is 28.

Exercise 3.3.1: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} 5 & 7 \\ 2 & 4 \end{bmatrix}, \qquad b = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (5)(4) - (7)(2) = 6.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 3 & 7 \\ 1 & 4 \end{bmatrix}, \qquad \det A_1(b) = 5.$$

The latter determinant is $(3)(4) - (7)(1) = 5$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix}, \qquad \det A_2(b) = -1.$$

The latter determinant is $(5)(1) - (3)(2) = -1$.

Divide and verify.

$$x = \begin{bmatrix} \frac{5}{6} \\ -\frac{1}{6} \end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 6. Substitution gives $Ax = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = b$, checking every original equation.

Exercise 3.3.2: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} 4 & 1 \\ 5 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 6 \\ 7 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (4)(2) - (1)(5) = 3.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 6 & 1 \\ 7 & 2 \end{bmatrix}, \qquad \det A_1(b) = 5.$$

The latter determinant is $(6)(2) - (1)(7) = 5$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} 4 & 6 \\ 5 & 7 \end{bmatrix}, \qquad \det A_2(b) = -2.$$

The latter determinant is $(4)(7) - (6)(5) = -2$.

Divide and verify.

$$x = \begin{bmatrix} \frac{5}{3} \\ -\frac{2}{3} \end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 3. Substitution gives $Ax = \begin{bmatrix} 6 \\ 7 \end{bmatrix} = b$, checking every original equation.

Exercise 3.3.3: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} 3 & -2 \\ -5 & 6 \end{bmatrix}, \qquad b = \begin{bmatrix} 7 \\ -5 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (3)(6) - (-2)(-5) = 8.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 7 & -2 \\ -5 & 6 \end{bmatrix}, \qquad \det A_1(b) = 32.$$

The latter determinant is $(7)(6) - (-2)(-5) = 32$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} 3 & 7 \\ -5 & -5 \end{bmatrix}, \qquad \det A_2(b) = 20.$$

The latter determinant is $(3)(-5) - (7)(-5) = 20$.

Divide and verify.

$$x = \begin{bmatrix} 4 \\ \frac{5}{2} \end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 8. Substitution gives $Ax = \begin{bmatrix} 7 \\ -5 \end{bmatrix} = b$, checking every original equation.

Exercise 3.3.4: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} -5 & 3 \\ 3 & -1 \end{bmatrix}, \qquad b = \begin{bmatrix} 9 \\ -5 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (-5)(-1) - (3)(3) = -4.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 9 & 3 \\ -5 & -1 \end{bmatrix}, \qquad \det A_1(b) = 6.$$

The latter determinant is $(9)(-1) - (3)(-5) = 6$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} -5 & 9 \\ 3 & -5 \end{bmatrix}, \qquad \det A_2(b) = -2.$$

The latter determinant is $(-5)(-5) - (9)(3) = -2$.

Divide and verify.

$$x = \begin{bmatrix} -\frac{3}{2} \\ \frac{1}{2} \end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by -4. Substitution gives $Ax = \begin{bmatrix} 9 \\ -5 \end{bmatrix} = b$, checking every original equation.

Exercise 3.3.5: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -3 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 7 \\ -8 \\ -3 \end{bmatrix}.$$

Check the denominator first.

$$\det A = (2)(-1)^{1+1} \begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} + (1)(-1)^{1+2} \begin{vmatrix} -3 & 1 \\ 0 & 2 \end{vmatrix} = 4.$$

The 2×2 determinants used above are

$$\det A_{11} = (0)(2) - (1)(1) = -1$$
$$\det A_{12} = (-3)(2) - (1)(0) = -6.$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b) = \begin{bmatrix} 7 & 1 & 0 \\ -8 & 0 & 1 \\ -3 & 1 & 2 \end{bmatrix}, \qquad \det A_1(b) = 6.$$

Its first-row cofactor expansion evaluates to $(7)(-1) + (1)(13) + (0)(-8) = 6$. Column 2 is replaced by b:

$$A_2(b) = \begin{bmatrix} 2 & 7 & 0 \\ -3 & -8 & 1 \\ 0 & -3 & 2 \end{bmatrix}, \qquad \det A_2(b) = 16.$$

Its first-row cofactor expansion evaluates to $(2)(-13) + (7)(6) + (0)(9) = 16$. Column 3 is replaced by b:

$$A_3(b) = \begin{bmatrix} 2 & 1 & 7 \\ -3 & 0 & -8 \\ 0 & 1 & -3 \end{bmatrix}, \qquad \det A_3(b) = -14.$$

Its first-row cofactor expansion evaluates to $(2)(8)+(1)(-9)+(7)(-3)=-14$.

Divide and verify.

$$x=\begin{bmatrix}\frac{3}{2}\\4\\-\frac{7}{2}\end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 4. Substitution gives $Ax=\begin{bmatrix}7\\-8\\-3\end{bmatrix}=b$, checking every original equation.

Exercise 3.3.6: Cramer's rule with all column replacements

What we must show. Solve the system using column replacements. Its matrix form is

$$A=\begin{bmatrix}2&1&1\\-1&0&2\\3&1&3\end{bmatrix},\qquad b=\begin{bmatrix}4\\2\\-2\end{bmatrix}.$$

Check the denominator first.

$$\det A=(2)(-1)^{1+1}\begin{vmatrix}0&2\\1&3\end{vmatrix}+(1)(-1)^{1+2}\begin{vmatrix}-1&2\\3&3\end{vmatrix}+(1)(-1)^{1+3}\begin{vmatrix}-1&0\\3&1\end{vmatrix}=4.$$

The 2×2 determinants used above are

$$\begin{aligned}\det A_{11}&=(0)(3)-(2)(1)=-2\\ \det A_{12}&=(-1)(3)-(2)(3)=-9\\ \det A_{13}&=(-1)(1)-(0)(3)=-1.\end{aligned}$$

The denominator is nonzero, so there is exactly one solution.

Replace each column in turn, keeping the others fixed.

Column 1 is replaced by b:

$$A_1(b)=\begin{bmatrix}4&1&1\\2&0&2\\-2&1&3\end{bmatrix},\qquad \det A_1(b)=-16.$$

Its first-row cofactor expansion evaluates to $(4)(-2)+(1)(-10)+(1)(2)=-16$. Column 2 is replaced by b:

$$A_2(b)=\begin{bmatrix}2&4&1\\-1&2&2\\3&-2&3\end{bmatrix},\qquad \det A_2(b)=52.$$

Its first-row cofactor expansion evaluates to $(2)(10)+(4)(9)+(1)(-4)=52$. Column 3 is replaced by b:

$$A_3(b)=\begin{bmatrix}2&1&4\\-1&0&2\\3&1&-2\end{bmatrix},\qquad \det A_3(b)=-4.$$

Its first-row cofactor expansion evaluates to $(2)(-2)+(1)(4)+(4)(-1)=-4$.

Divide and verify.

$$x=\begin{bmatrix}-4\\13\\-1\end{bmatrix}.$$

Each component is the corresponding replacement determinant divided by 4. Substitution gives $Ax=\begin{bmatrix}4\\2\\-2\end{bmatrix}=b$, checking every original equation.

Exercise 3.3.7: A parameter and the nonzero-denominator condition

What we must show. Determine exactly when the parameterized system has a unique solution.

$$A=\begin{bmatrix}6s&4\\9&2s\end{bmatrix},\qquad b=\begin{bmatrix}5\\-2\end{bmatrix}.$$

Its determinant is

$$\det A=12\left(s^2-3\right).$$

A square system has a unique solution precisely when its coefficient matrix is invertible, so the required parameter condition is $s\neq\sqrt{3},-\sqrt{3}$. Under that condition the two replacement determinants are

$$\det A_1(b)=10s+8,\qquad \det A_2(b)=-12s-45.$$

Cramer's rule therefore gives

$$x_1=\frac{5s+4}{6\left(s^2-3\right)},\qquad x_2=-\frac{4s+15}{4\left(s^2-3\right)}.$$

These formulas were checked by multiplying by A: the result is exactly b wherever the denominator is nonzero. At an excluded parameter the matrix is singular, so the system cannot have a unique solution; division by its zero determinant is not allowed.

Exercise 3.3.8: A parameter and the nonzero-denominator condition

What we must show. Determine exactly when the parameterized system has a unique solution.

$$A=\begin{bmatrix}3s&-5\\9&5s\end{bmatrix},\qquad b=\begin{bmatrix}3\\2\end{bmatrix}.$$

Its determinant is

$$\det A=15\left(s^2+3\right).$$

A square system has a unique solution precisely when its coefficient matrix is invertible, so the required parameter condition is every real s. Under that condition the two replacement determinants are

$$\det A_1(b) = 15s + 10, \qquad \det A_2(b) = 6s - 27.$$

Cramer's rule therefore gives

$$x_1 = \frac{3s + 2}{3\left(s^2 + 3\right)}, \qquad x_2 = \frac{2s - 9}{5\left(s^2 + 3\right)}.$$

These formulas were checked by multiplying by A: the result is exactly b wherever the denominator is nonzero. At an excluded parameter the matrix is singular, so the system cannot have a unique solution; division by its zero determinant is not allowed.

Exercise 3.3.9: A parameter and the nonzero-denominator condition

What we must show. Determine exactly when the parameterized system has a unique solution.

$$A = \begin{bmatrix} s & -2s \\ 3 & 6s \end{bmatrix}, \qquad b = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

Its determinant is

$$\det A = 6s\left(s + 1\right).$$

A square system has a unique solution precisely when its coefficient matrix is invertible, so the required parameter condition is $s \neq 0, -1$. Under that condition the two replacement determinants are

$$\det A_1(b) = 2s, \qquad \det A_2(b) = 4s + 3.$$

Cramer's rule therefore gives

$$x_1 = \frac{1}{3\left(s + 1\right)}, \qquad x_2 = \frac{4s + 3}{6s\left(s + 1\right)}.$$

These formulas were checked by multiplying by A: the result is exactly b wherever the denominator is nonzero. At an excluded parameter the matrix is singular, so the system cannot have a unique solution; division by its zero determinant is not allowed.

Exercise 3.3.10: A parameter and the nonzero-denominator condition

What we must show. Determine exactly when the parameterized system has a unique solution.

$$A = \begin{bmatrix} 2s & 1 \\ 3s & 6s \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Its determinant is

$$\det A = 3s\left(4s - 1\right).$$

A square system has a unique solution precisely when its coefficient matrix is invertible, so the required parameter condition is $s \neq 0, 1/4$. Under that condition the two replacement determinants are

$$\det A_1(b) = 6s - 2, \qquad \det A_2(b) = s.$$

Cramer's rule therefore gives

$$x_1 = \frac{2\,(3s-1)}{3s\,(4s-1)}, \qquad x_2 = \frac{1}{3\,(4s-1)}.$$

These formulas were checked by multiplying by A: the result is exactly b wherever the denominator is nonzero. At an excluded parameter the matrix is singular, so the system cannot have a unique solution; division by its zero determinant is not allowed.

Exercise 3.3.11: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 0 & -2 & -1 \\ 3 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned} C_{11} &= (-1)^{1+1}[(0)(1) - (0)(1)] = 0 \\ C_{12} &= (-1)^{1+2}[(3)(1) - (0)(-1)] = -3 \\ C_{13} &= (-1)^{1+3}[(3)(1) - (0)(-1)] = 3 \\ C_{21} &= (-1)^{2+1}[(-2)(1) - (-1)(1)] = 1 \\ C_{22} &= (-1)^{2+2}[(0)(1) - (-1)(-1)] = -1 \\ C_{23} &= (-1)^{2+3}[(0)(1) - (-2)(-1)] = 2 \\ C_{31} &= (-1)^{3+1}[(-2)(0) - (-1)(0)] = 0 \\ C_{32} &= (-1)^{3+2}[(0)(0) - (-1)(3)] = -3 \\ C_{33} &= (-1)^{3+3}[(0)(0) - (-2)(3)] = 6 \end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} 0 & -3 & 3 \\ 1 & -1 & 2 \\ 0 & -3 & 6 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} 0 & 1 & 0 \\ -3 & -1 & -3 \\ 3 & 2 & 6 \end{bmatrix}.$$

The first-row expansion gives $\det A = (0)(0) + (-2)(-3) + (-1)(3) = 3 \neq 0$. Thus

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 0 & 1 & 0 \\ -3 & -1 & -3 \\ 3 & 2 & 6 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = 3I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.12: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 2 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned} C_{11} &= (-1)^{1+1}[(-2)(0) - (1)(1)] = -1 \\ C_{12} &= (-1)^{1+2}[(2)(0) - (1)(0)] = 0 \\ C_{13} &= (-1)^{1+3}[(2)(1) - (-2)(0)] = 2 \\ C_{21} &= (-1)^{2+1}[(1)(0) - (3)(1)] = 3 \\ C_{22} &= (-1)^{2+2}[(1)(0) - (3)(0)] = 0 \\ C_{23} &= (-1)^{2+3}[(1)(1) - (1)(0)] = -1 \\ C_{31} &= (-1)^{3+1}[(1)(1) - (3)(-2)] = 7 \\ C_{32} &= (-1)^{3+2}[(1)(1) - (3)(2)] = 5 \\ C_{33} &= (-1)^{3+3}[(1)(-2) - (1)(2)] = -4 \end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} -1 & 0 & 2 \\ 3 & 0 & -1 \\ 7 & 5 & -4 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} -1 & 3 & 7 \\ 0 & 0 & 5 \\ 2 & -1 & -4 \end{bmatrix}.$$

The first-row expansion gives $\det A = (1)(-1) + (1)(0) + (3)(2) = 5 \neq 0$. Thus

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -1 & 3 & 7 \\ 0 & 0 & 5 \\ 2 & -1 & -4 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = 5I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.13: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 3 & 5 & 4 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned}
C_{11} &= (-1)^{1+1}[(0)(1) - (1)(1)] = -1 \\
C_{12} &= (-1)^{1+2}[(1)(1) - (1)(2)] = 1 \\
C_{13} &= (-1)^{1+3}[(1)(1) - (0)(2)] = 1 \\
C_{21} &= (-1)^{2+1}[(5)(1) - (4)(1)] = -1 \\
C_{22} &= (-1)^{2+2}[(3)(1) - (4)(2)] = -5 \\
C_{23} &= (-1)^{2+3}[(3)(1) - (5)(2)] = 7 \\
C_{31} &= (-1)^{3+1}[(5)(1) - (4)(0)] = 5 \\
C_{32} &= (-1)^{3+2}[(3)(1) - (4)(1)] = 1 \\
C_{33} &= (-1)^{3+3}[(3)(0) - (5)(1)] = -5
\end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} -1 & 1 & 1 \\ -1 & -5 & 7 \\ 5 & 1 & -5 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} -1 & -1 & 5 \\ 1 & -5 & 1 \\ 1 & 7 & -5 \end{bmatrix}.$$

The first-row expansion gives $\det A = (3)(-1) + (5)(1) + (4)(1) = 6 \neq 0$. Thus

$$A^{-1} = \frac{1}{6} \begin{bmatrix} -1 & -1 & 5 \\ 1 & -5 & 1 \\ 1 & 7 & -5 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = 6I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.14: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 3 & 6 & 7 \\ 0 & 2 & 1 \\ 2 & 3 & 4 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned}
C_{11} &= (-1)^{1+1}[(2)(4) - (1)(3)] = 5 \\
C_{12} &= (-1)^{1+2}[(0)(4) - (1)(2)] = 2 \\
C_{13} &= (-1)^{1+3}[(0)(3) - (2)(2)] = -4 \\
C_{21} &= (-1)^{2+1}[(6)(4) - (7)(3)] = -3 \\
C_{22} &= (-1)^{2+2}[(3)(4) - (7)(2)] = -2 \\
C_{23} &= (-1)^{2+3}[(3)(3) - (6)(2)] = 3 \\
C_{31} &= (-1)^{3+1}[(6)(1) - (7)(2)] = -8 \\
C_{32} &= (-1)^{3+2}[(3)(1) - (7)(0)] = -3 \\
C_{33} &= (-1)^{3+3}[(3)(2) - (6)(0)] = 6
\end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} 5 & 2 & -4 \\ -3 & -2 & 3 \\ -8 & -3 & 6 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} 5 & -3 & -8 \\ 2 & -2 & -3 \\ -4 & 3 & 6 \end{bmatrix}.$$

The first-row expansion gives $\det A = (3)(5) + (6)(2) + (7)(-4) = -1 \neq 0$. Thus

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 5 & -3 & -8 \\ 2 & -2 & -3 \\ -4 & 3 & 6 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = -1 I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.15: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 3 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 3 & 2 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$C_{11} = (-1)^{1+1}[(1)(2) - (0)(3)] = 2$$
$$C_{12} = (-1)^{1+2}[(-1)(2) - (0)(-2)] = 2$$
$$C_{13} = (-1)^{1+3}[(-1)(3) - (1)(-2)] = -1$$
$$C_{21} = (-1)^{2+1}[(0)(2) - (0)(3)] = 0$$
$$C_{22} = (-1)^{2+2}[(3)(2) - (0)(-2)] = 6$$
$$C_{23} = (-1)^{2+3}[(3)(3) - (0)(-2)] = -9$$
$$C_{31} = (-1)^{3+1}[(0)(0) - (0)(1)] = 0$$
$$C_{32} = (-1)^{3+2}[(3)(0) - (0)(-1)] = 0$$
$$C_{33} = (-1)^{3+3}[(3)(1) - (0)(-1)] = 3$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} 2 & 2 & -1 \\ 0 & 6 & -9 \\ 0 & 0 & 3 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 6 & 0 \\ -1 & -9 & 3 \end{bmatrix}.$$

The first-row expansion gives $\det A = (3)(2) + (0)(2) + (0)(-1) = 6 \neq 0$. Thus

$$A^{-1} = \frac{1}{6} \begin{bmatrix} 2 & 0 & 0 \\ 2 & 6 & 0 \\ -1 & -9 & 3 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = 6I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.16: All cofactors, the adjugate, and the inverse

What we must show. Find the adjugate and inverse of

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 0 & -3 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$$

Compute all signed minors.

Deleting row i and column j and multiplying its 2×2 determinant by $(-1)^{i+j}$ gives:

$$\begin{aligned}
C_{11} &= (-1)^{1+1}[(-3)(3) - (1)(0)] = -9 \\
C_{12} &= (-1)^{1+2}[(0)(3) - (1)(0)] = 0 \\
C_{13} &= (-1)^{1+3}[(0)(0) - (-3)(0)] = 0 \\
C_{21} &= (-1)^{2+1}[(2)(3) - (4)(0)] = -6 \\
C_{22} &= (-1)^{2+2}[(1)(3) - (4)(0)] = 3 \\
C_{23} &= (-1)^{2+3}[(1)(0) - (2)(0)] = 0 \\
C_{31} &= (-1)^{3+1}[(2)(1) - (4)(-3)] = 14 \\
C_{32} &= (-1)^{3+2}[(1)(1) - (4)(0)] = -1 \\
C_{33} &= (-1)^{3+3}[(1)(-3) - (2)(0)] = -3
\end{aligned}$$

Transpose, rather than using the cofactor matrix directly.

$$C = \begin{bmatrix} -9 & 0 & 0 \\ -6 & 3 & 0 \\ 14 & -1 & -3 \end{bmatrix}, \qquad \operatorname{adj} A = C^T = \begin{bmatrix} -9 & -6 & 14 \\ 0 & 3 & -1 \\ 0 & 0 & -3 \end{bmatrix}.$$

The first-row expansion gives $\det A = (1)(-9) + (2)(0) + (4)(0) = -9 \neq 0$. Thus

$$A^{-1} = \frac{1}{-9} \begin{bmatrix} -9 & -6 & 14 \\ 0 & 3 & -1 \\ 0 & 0 & -3 \end{bmatrix}.$$

As an exact check, multiplication gives $A \operatorname{adj} A = -9I_3$. Dividing this identity by the nonzero determinant gives $AA^{-1} = I_3$.

Exercise 3.3.17: The adjugate formula and its consequences

For $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, the four cofactors are $d, -c, -b, a$, in row order. Transposing gives

$$\operatorname{adj} A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Theorem 8 therefore gives $A^{-1} = (ad - bc)^{-1} \left[\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right]$ when $ad - bc \neq 0$. This is exactly the earlier 2×2 inverse formula. The nonzero-denominator condition is part of both formulas.

Exercise 3.3.18: The adjugate formula and its consequences

Each minor matrix has integer entries. Its determinant is an integer because a determinant is a finite sum of signed products of its entries. Cofactors, and hence every entry of the adjugate, are therefore integers. Since $\det A = 1$, Theorem 8 gives $A^{-1} = \operatorname{adj} A/1$, whose entries are all integers. The hypothesis $\det A = 1$ is essential to this simple argument; division by a general nonzero integer need not leave integers.

Exercise 3.3.19: Area from translated adjacent edges

Translate the vertex $p = (0,\ 0)$ to the origin. The two adjacent edges are

$$u = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \qquad v = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$

The remaining listed vertex is $p+u+v = (11,\ 6)$, so these really are adjacent edges of the parallelogram, rather than a mistaken choice of diagonals. Translation preserves area, giving

$$\text{area} = \left\| \begin{vmatrix} 5 & 6 \\ 2 & 4 \end{vmatrix} \right\| = |8| = 8.$$

Exercise 3.3.20: Area from translated adjacent edges

Translate the vertex $p = (0,\ 0)$ to the origin. The two adjacent edges are

$$u = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \qquad v = \begin{bmatrix} 4 \\ -5 \end{bmatrix}.$$

The remaining listed vertex is $p + u + v = (3,\ -2)$, so these really are adjacent edges of the parallelogram, rather than a mistaken choice of diagonals. Translation preserves area, giving

$$\text{area} = \left\| \begin{vmatrix} -1 & 4 \\ 3 & -5 \end{vmatrix} \right\| = |-7| = 7.$$

Exercise 3.3.21: Area from translated adjacent edges

Translate the vertex $p = (-1,\ 0)$ to the origin. The two adjacent edges are

$$u = \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \qquad v = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

The remaining listed vertex is $p+u+v = (2,\ 1)$, so these really are adjacent edges of the parallelogram, rather than a mistaken choice of diagonals. Translation preserves area, giving

$$\text{area} = \left\| \begin{vmatrix} 1 & 2 \\ 5 & -4 \end{vmatrix} \right\| = |-14| = 14.$$

Exercise 3.3.22: Area from translated adjacent edges

Translate the vertex $p = (0,\ -2)$ to the origin. The two adjacent edges are

$$u = \begin{bmatrix} 6 \\ 1 \end{bmatrix}, \qquad v = \begin{bmatrix} -3 \\ 3 \end{bmatrix}.$$

The remaining listed vertex is $p+u+v = (3,\ 2)$, so these really are adjacent edges of the parallelogram, rather than a mistaken choice of diagonals. Translation preserves area, giving

$$\text{area} = \left\| \begin{vmatrix} 6 & -3 \\ 1 & 3 \end{vmatrix} \right\| = |21| = 21.$$

Exercise 3.3.23: Volume of a parallelepiped

Place the three adjacent edge vectors in columns:

$$A = \begin{bmatrix} 1 & 1 & 7 \\ 0 & 2 & 1 \\ -2 & 4 & 0 \end{bmatrix}.$$

$$\det A = (1)(-1)^{1+1}\begin{vmatrix} 2 & 1 \\ 4 & 0 \end{vmatrix} + (1)(-1)^{1+2}\begin{vmatrix} 0 & 1 \\ -2 & 0 \end{vmatrix} + (7)(-1)^{1+3}\begin{vmatrix} 0 & 2 \\ -2 & 4 \end{vmatrix} = 22.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (2)(0) - (1)(4) = -4 \\ \det A_{12} &= (0)(0) - (1)(-2) = 2 \\ \det A_{13} &= (0)(4) - (2)(-2) = 4. \end{aligned}$$

The determinant is signed, so the ordinary volume is $|22| = 22$. If the columns are reordered, the signed determinant may change but this volume does not.

Exercise 3.3.24: Volume of a parallelepiped

Place the three adjacent edge vectors in columns:

$$A = \begin{bmatrix} 1 & -2 & -1 \\ 4 & -5 & 2 \\ 0 & 2 & -1 \end{bmatrix}.$$

$$\det A = (1)(-1)^{1+1}\begin{vmatrix} -5 & 2 \\ 2 & -1 \end{vmatrix} + (-2)(-1)^{1+2}\begin{vmatrix} 4 & 2 \\ 0 & -1 \end{vmatrix} + (-1)(-1)^{1+3}\begin{vmatrix} 4 & -5 \\ 0 & 2 \end{vmatrix} = -15.$$

The 2×2 determinants used above are

$$\begin{aligned} \det A_{11} &= (-5)(-1) - (2)(2) = 1 \\ \det A_{12} &= (4)(-1) - (2)(0) = -4 \\ \det A_{13} &= (4)(2) - (-5)(0) = 8. \end{aligned}$$

The determinant is signed, so the ordinary volume is $|-15| = 15$. If the columns are reordered, the signed determinant may change but this volume does not.

Exercise 3.3.25: Geometry and linearity

Let the columns be a_1, a_2, a_3. Their parallelepiped has volume $|\det A|$. If they are linearly dependent, all three vectors lie in a plane through the origin (or a smaller subspace), so the parallelepiped is flat and has zero volume. Conversely, three independent vectors form a genuinely three-dimensional parallelepiped: the first two span a base plane of positive area, and the third has positive distance from that plane, giving positive base-area times height. Thus volume is zero exactly when the columns are dependent. By the Invertible Matrix Theorem, dependence is exactly noninvertibility. This geometric proof does not invoke the determinant criterion in Section 3.2, Theorem 4.

Exercise 3.3.26: Geometry and linearity

What we must show. Prove equality of the two sets, which requires both inclusions. A point y belongs to $T(p+S)$ exactly when $y = T(p+s)$ for some $s \in S$. Linearity changes this to $y = T(p)+T(s)$ for some $s \in S$, which is exactly membership in $T(p) + T(S)$. Every step can be reversed, so

$$T(p + S) = T(p) + T(S).$$

The argument also handles $S = \varnothing$, when both sets are empty. A linear map sends a translated set to the corresponding translated image.

Exercise 3.3.27: Area before and after a linear map

The original edge matrix and its image are

$$B = \begin{bmatrix} -2 & -2 \\ 3 & 5 \end{bmatrix}, \qquad AB = \begin{bmatrix} -18 & -22 \\ 12 & 16 \end{bmatrix}.$$

The original area is $|\det B| = |-4| = 4$. Since $\det A = 6$, the image area is

$$|\det A|\,|\det B| = 6 \cdot 4 = 24.$$

A direct determinant of the displayed image edge matrix gives -24, confirming the same absolute area.

Exercise 3.3.28: Area before and after a linear map

The original edge matrix and its image are

$$B = \begin{bmatrix} 4 & 0 \\ -7 & 1 \end{bmatrix}, \qquad AB = \begin{bmatrix} 14 & 2 \\ -3 & 1 \end{bmatrix}.$$

The original area is $|\det B| = |4| = 4$. Since $\det A = 5$, the image area is

$$|\det A|\,|\det B| = 5 \cdot 4 = 20.$$

A direct determinant of the displayed image edge matrix gives 20, confirming the same absolute area.

Exercise 3.3.29: Area and volume formulas

The diagonal from 0 to $v_1 + v_2$ divides the parallelogram into two congruent triangles. Each has half the parallelogram's area, so the triangle with vertices $0, v_1, v_2$ has area

$$\frac{1}{2}|\det[v_1\ v_2]|.$$

If the two edges are dependent, the parallelogram and triangle are both flat and the same formula gives zero.

Exercise 3.3.30: Area and volume formulas

What we must show. Derive an orientation-independent triangle-area formula.

Translate one vertex to the origin.

Subtract (x_1, y_1) from all vertices. The new edge vectors are $u = (x_2 - x_1, y_2 - y_1)$ and $v = (x_3 - x_1, y_3 - y_1)$. Translation does not change area, so Exercise 29 gives $\frac{1}{2}|\det[u\ v]|$.

Connect this to the displayed determinant.

In the 3×3 matrix in the exercise, subtract row 1 from rows 2 and 3. Expanding down column 3 then gives

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix}.$$

This last matrix is $[u\ v]^T$, with the same determinant as $[u\ v]$.

Conclusion. The ordinary area is one half the *absolute value* of the displayed 3×3 determinant. The supplied print formula omits the absolute-value bars; without them it is a signed area and can be negative. For vertices $(0,0), (0,1), (1,0)$ the displayed determinant is -1, so the omission cannot be valid for ordinary area with arbitrary vertex order.

Exercise 3.3.31: Area and volume formulas

(a) Describe the image in both directions.

The map is $T(u_1, u_2, u_3) = (au_1, bu_2, cu_3)$. Because $a, b, c > 0$, it has inverse $u = (x_1/a, x_2/b, x_3/c)$. Thus

$$u_1^2 + u_2^2 + u_3^2 \le 1 \quad \Longleftrightarrow \quad \frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} \le 1.$$

So the ball maps exactly onto the solid ellipsoid, and equality describes its boundary.

(b) Multiply the original volume by the determinant factor.

The diagonal matrix has determinant $abc > 0$. Hence the image volume is $abc(4\pi/3) = 4\pi abc/3$.

Exercise 3.3.32: Area and volume formulas

(a) Specify the map by its columns.

Take $A = [v_1\ v_2\ v_3]$ and $T(x) = Ax$. Then $T(e_i) = v_i$ and $T(0) = 0$. More explicitly, the standard tetrahedron consists of the points $t_1e_1 + t_2e_2 + t_3e_3$ with $t_i \ge 0$ and $t_1 + t_2 + t_3 \le 1$. Their images are exactly the same combinations of v_1, v_2, v_3, namely S'.

(b) Compute the scale factor from a unit reference volume.

The standard base triangle in the coordinate plane has area $1/2$, and the perpendicular height to e_3 is 1. Thus $\text{vol}(S) = (1/3)(1/2)(1) = 1/6$. The linear map multiplies volume by $|\det A|$, giving

$$\text{vol}(S') = \frac{1}{6}|\det[v_1\ v_2\ v_3]|.$$

The same formula gives zero when the three vectors are dependent and the image tetrahedron is flat.

Exercise 3.3.33: A floating-point adjugate experiment

Use the reproducible random integer matrix

$$A = \begin{bmatrix} -3 & 8 & -2 & 4 \\ 7 & 8 & 9 & -1 \\ 3 & 8 & -9 & -3 \\ 0 & 4 & -5 & -1 \end{bmatrix}.$$

The following computation forms every cofactor independently in double precision, transposes, divides by the determinant, and compares with a numerical inverse.

```
import numpy as np
A = np.array([[-3, 8, -2, 4],
              [7, 8, 9, -1],
              [3, 8, -9, -3],
              [0, 4, -5, -1]],float)
C = np.zeros((4,4))
for i in range(4):
    for j in range(4):
        minor = np.delete(np.delete(A,i,0),j,1)
        C[i,j] = (-1)**(i+j)*np.linalg.det(minor)
B = C.T/np.linalg.det(A)
print(np.max(np.abs(B-np.linalg.inv(A))))
```

In this run the maximum absolute entry of $B - \mathrm{inv}(A)$ was approximately 1.55×10^{-15}. The exact determinant is $-640 \neq 0$, and exact rational arithmetic also gives $A \operatorname{adj} A = (\det A)I$. The floating discrepancy is roundoff, not a failure of the inverse identity; a different numerical library may change the last digits. Double precision stores about 16 decimal significant digits, not infinitely many, so we report a maximum error rather than an unjustified assertion of bit-for-bit equality.

Exercise 3.3.34: Cramer's rule and an inverse on the same random system

For one reproducible random system, take

$$A = \begin{bmatrix} 4 & 6 & 0 & -7 \\ 3 & -6 & 9 & -3 \\ -1 & 6 & -5 & -5 \\ -2 & 0 & 9 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} -7 \\ 9 \\ -2 \\ -5 \end{bmatrix}.$$

The denominator and four column-replacement determinants are

$$\det A = -2448, \qquad (\det A_1, \det A_2, \det A_3, \det A_4) = (2616,\ 5067,\ 1188,\ 3390)\,.$$

Hence Cramer's rule gives

$$x = \begin{bmatrix} -\frac{109}{102} \\ -\frac{563}{272} \\ -\frac{33}{68} \\ -\frac{565}{408} \end{bmatrix}.$$

Direct exact multiplication gives $Ax = b$, and exact inversion gives the identical vector $A^{-1}b$. Here is a matrix-program command for the second component, with Python's zero-based column indexing:

```
from sympy import Matrix
A = Matrix([[4, 6, 0, -7], [3, -6, 9, -3], [-1, 6, -5, -5], [-2, 0, 9, 2]])
b = Matrix([-7, 9, -2, -5])
A2 = A.copy()
A2[:,1] = b
x2 = A2.det()/A.det()
assert x2 == (A.inv()*b)[1]
```

In MATLAB the corresponding column-replacement command is

```
A2 = A; A2(:,2) = b; x2 = det(A2)/det(A);
```

The determinant must be checked nonzero before division; a singular random draw would require a new nonsingular test matrix.

Exercise 3.3.35: Count arithmetic instead of confusing timing with flops

What we must show. Compare operation counts for a 30×30 inverse and an adjugate-based inverse. The question is conditional on having the historical MATLAB `flops` command. Modern MATLAB and NumPy do not provide that same counter, so a timing measurement must not be called a flop count. Instead we give an explicit arithmetic-count model that is reproducible and makes the comparison precise.

Count each scalar addition, subtraction, multiplication, or division as one flop; exclude comparisons, indexing, and memory movement. Assume a dense nonsingular matrix with no zero shortcuts. For elimination at step k, with $r = n - k - 1$ rows remaining, computing multipliers costs r divisions and updating the trailing square costs $2r^2$ flops. Thus

$$L(n) = \sum_{r=1}^{n-1} (r + 2r^2).$$

After one LU factorization, solve n right-hand sides to find the inverse. Unit-lower-triangular forward substitution costs $n(n-1)$ per right-hand side; back substitution costs $n(n-1) + n$. Hence the total is

$$L(n) + n(2n^2 - n).$$

For the cofactor method, form all n^2 determinants of size $n - 1$, each with elimination cost $L(n-1)$ plus $n - 2$ diagonal-product multiplications. Form the original determinant with cost $L(n) + n - 1$, then divide n^2 adjugate entries by it. The cost is

$$n^2[L(n-1) + n - 2] + L(n) + n - 1 + n^2.$$

At $n = 30$, these formulas give 70645 flops for LU plus solves, versus 14294274 for the independently evaluated adjugate formula, about 202.3 times as many. They can be checked by

```
def L(n):
```

```
    return sum(r+2*r*r for r in range(1,n))
n = 30
print(L(n)+n*(2*n*n-n))
print(n*n*(L(n-1)+n-2)+L(n)+n-1+n*n)
```

These are model counts, not measurements from an unavailable old MATLAB counter. Implementations that reuse subdeterminants or use different algorithms have different counts. The main conclusion is that forming all cofactors independently is an $O(n^5)$ method here, while factorization and solves are $O(n^3)$. Cofactor *expansion* without elimination would be still more expensive.

3.S Chapter 3 supplementary exercises

These exercises combine determinant calculations with proofs. A counterexample disproves a universal claim; several successful numerical experiments do not prove one. Throughout this supplement matrices and scalars are real unless the problem says otherwise. It is helpful to keep three facts distinct: row replacement preserves a determinant, scaling *one* row scales it once, and scaling an entire $n \times n$ matrix scales it n times.

Exercise 3.S.1: Sixteen statements with explanations

What we must show. Judge all sixteen statements, giving a reason rather than just a letter.

(a) **True.** Zero determinant means dependent columns. For two columns u, v, dependence makes one a multiple of the other: if a dependence $cu + dv = 0$ has $c \neq 0$, solve for u; otherwise $d \neq 0$ gives $v = 0$, a multiple of u. This includes zero columns.

(b) **True.** Interchanging the equal rows leaves the matrix unchanged but reverses the determinant, so $d = -d$ and $d = 0$.

(c) **False.** Scaling every row of a 3×3 matrix gives $\det(5A) = 5^3 \det A$, not $5 \det A$. For $A = I_3$, the two proposed sides are 125 and 5.

(d) **False.** For $n = 2$, take $A = \operatorname{diag}(2, 1)$ and $B = \operatorname{diag}(3, 1)$. Their determinants are 2 and 3, but $\det(A + B) = \det \operatorname{diag}(5, 2) = 10$, not 5.

(e) **False.** Multiplicativity gives $\det(A^3) = (\det A)^3 = 2^3 = 8$, not 6.

(f) **False.** A single interchange gives $\det B = -\det A$. Interchanging two rows of I_2 gives determinant -1, not 1.

(g) **True.** Scaling row 3 by 5 multiplies the determinant by 5, with all other rows fixed.

(h) **True.** Add the stated multiples one at a time. Each is a row replacement using an unchanged other row, so each preserves the determinant and their finite succession does too.

(i) **False.** Transposition preserves the determinant. The identity matrix disproves a general minus sign.

(j) **False in general.** The correct rule is $\det(-A) = (-1)^n \det A$. For I_2 this is 1, not -1; the proposed rule holds in odd dimensions.

(k) **True.** Multiplicativity and transpose invariance give $\det(A^T A) = (\det A)^2 \geq 0$.

(l) **False.** Cramer's quotient formulas require a nonzero coefficient determinant. The system $0x = 1$, for example, is inconsistent and cannot be solved by dividing replacement determinants by 0.

(m) **False.** The parallelogram has area 10; the stated triangle has half that area, namely 5.

(n) **True.** Taking determinants in $A^3 = 0$ yields $(\det A)^3 = 0$, hence $\det A = 0$.

(o) **False.** The correct formula is $\det(A^{-1}) = 1/\det A$. Taking $A = [2]$ gives $1/2$, not 2.

(p) **True.** Apply multiplicativity to $AA^{-1} = I$: the two determinant factors have product 1.

Exercise 3.S.2: Equally spaced rows

For the matrix with rows $(12, 13, 14)$, $(15, 16, 17)$, and $(18, 19, 20)$, perform $R_3 \leftarrow R_3 - R_2$ first, giving $(3, 3, 3)$ in row 3. Next do $R_2 \leftarrow R_2 - R_1$, also giving $(3, 3, 3)$. Finally $R_3 \leftarrow R_3 - R_2$ creates a zero row. Only row replacements were used, so the determinant is unchanged and equals zero.

Exercise 3.S.3: A symbolic determinant that vanishes

Begin with rows $(1, a, b+c)$, $(1, b, a+c)$, and $(1, c, a+b)$. Subtract row 1 from rows 2 and 3 to obtain

$$\begin{bmatrix} 1 & a & b+c \\ 0 & b-a & a-b \\ 0 & c-a & a-c \end{bmatrix}.$$

If $b = a$, row 2 is already zero. Otherwise replace row 3 by row 3 minus $\frac{c-a}{b-a}$ times row 2. Its second entry becomes zero, and its third becomes

$$(a-c) - \frac{c-a}{b-a}(a-b) = (a-c) + (c-a) = 0.$$

Thus a determinant-preserving operation makes a zero row in every case, proving the determinant is zero. The case split avoids dividing by a possibly zero parameter.

Exercise 3.S.4: Rows translated in a common direction

Subtract row 1 from rows 2 and 3. The matrix becomes

$$\begin{bmatrix} a & b & c \\ x & x & x \\ y & y & y \end{bmatrix}.$$

If $x = 0$, it has a zero row. If $x \neq 0$, replace row 3 by row 3 minus (y/x) times row 2; that gives a zero row. All operations were row replacements, so the original determinant is zero, including the special parameter cases.

Exercise 3.S.5: Choose sparse expansions

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 9 & 1 & 9 & 9 & 9 \\ 9 & 0 & 9 & 9 & 2 \\ 4 & 0 & 0 & 5 & 0 \\ 9 & 0 & 3 & 9 & 0 \\ 6 & 0 & 0 & 7 & 0 \end{vmatrix} = (1)(-1)^{1+2} D_2 = -12.$$

$$D_2 = \begin{vmatrix} 9 & 9 & 9 & 2 \\ 4 & 0 & 5 & 0 \\ 9 & 3 & 9 & 0 \\ 6 & 0 & 7 & 0 \end{vmatrix} = (2)(-1)^{1+4} D_3 = 12.$$

$$D_3 = \begin{vmatrix} 4 & 0 & 5 \\ 9 & 3 & 9 \\ 6 & 0 & 7 \end{vmatrix} = (3)(-1)^{2+2} D_4 = -6.$$

$$D_4 = \begin{vmatrix} 4 & 5 \\ 6 & 7 \end{vmatrix} = (4)(7) - (5)(6) = -2.$$

Thus the answer is -12.

Exercise 3.S.6: Choose sparse expansions

Choose a row or column with the most zeros at each stage. In the following calculation tree, D_1 is the required determinant; each other D_j is a smaller determinant. The exponent of -1 records the deleted row and column in that particular matrix.

$$D_1 = \begin{vmatrix} 4 & 8 & 8 & 8 & 5 \\ 0 & 1 & 0 & 0 & 0 \\ 6 & 8 & 8 & 8 & 7 \\ 0 & 8 & 8 & 3 & 0 \\ 0 & 8 & 2 & 0 & 0 \end{vmatrix} = (1)(-1)^{2+2} D_2 = 12.$$

$$D_2 = \begin{vmatrix} 4 & 8 & 8 & 5 \\ 6 & 8 & 8 & 7 \\ 0 & 8 & 3 & 0 \\ 0 & 2 & 0 & 0 \end{vmatrix} = (2)(-1)^{4+2} D_3 = 12.$$

$$D_3 = \begin{vmatrix} 4 & 8 & 5 \\ 6 & 8 & 7 \\ 0 & 3 & 0 \end{vmatrix} = (3)(-1)^{3+2} D_4 = 6.$$

$$D_4 = \begin{vmatrix} 4 & 5 \\ 6 & 7 \end{vmatrix} = (4)(7) - (5)(6) = -2.$$

Thus the answer is 12.

Exercise 3.S.7: Geometric and structural determinant proofs

Subtract row 2 from rows 1 and 3 in the displayed determinant. Expanding down its first column shows that its zero equation is equivalent to

$$(x - x_1)(y_2 - y_1) - (y - y_1)(x_2 - x_1) = 0.$$

This says the displacement $(x - x_1, y - y_1)$ is parallel to the nonzero direction $(x_2 - x_1, y_2 - y_1)$. To check without a geometric shortcut, if $x_2 \neq x_1$ solve for $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$; if $x_2 = x_1$, distinctness gives $y_2 \neq y_1$, and the equation reduces to $x = x_1$. These are exactly the nonvertical and vertical lines through the two distinct points.

Exercise 3.S.8: Geometric and structural determinant proofs

A direction vector for a line of finite slope m is $(1, m)$. Thus one valid determinant equation is

$$\begin{vmatrix} 1 & x & y \\ 1 & x_1 & y_1 \\ 0 & 1 & m \end{vmatrix} = 0.$$

Subtract row 2 from row 1, then expand down column 1. The left side is $-[(x - x_1)m - (y - y_1)]$, so setting it to zero is precisely $y - y_1 = m(x - x_1)$. Alternatively the third row can be $(1, x_1 + 1, y_1 + m)$, corresponding to a second point on the same line; subtracting row 2 converts it to the displayed version.

Exercise 3.S.9: Geometric and structural determinant proofs

Subtract row 1 from rows 2 and 3 in T. The two new rows are $(0, b - a, b^2 - a^2)$ and $(0, c - a, c^2 - a^2)$. Factor $b^2 - a^2 = (b - a)(b + a)$ and $c^2 - a^2 = (c - a)(c + a)$ to obtain

$$\begin{aligned} \det T &= (b - a)(c - a) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b + a \\ 0 & 1 & c + a \end{vmatrix} \\ &= (b - a)(c - a)[(c + a) - (b + a)] \\ &= (b - a)(c - a)(c - b). \end{aligned}$$

Row factoring is a polynomial identity, so no division by $b - a$ or $c - a$ was used; the formula remains valid when two parameters coincide.

Exercise 3.S.10: Geometric and structural determinant proofs

Expand along the first row of $V(t)$. Its cofactors depend only on the fixed numbers x_1, x_2, x_3, so the expansion is a polynomial in t of degree at most 3. The coefficient of t^3 is the cofactor in position $(1, 4)$:

$$-\begin{vmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{vmatrix} = -(x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

It is nonzero because the three numbers are distinct. Hence the degree is exactly 3. At $t = x_i$, the first row equals row $i + 1$, so the determinant is zero. Thus three graph points are $(x_1, 0)$, $(x_2, 0)$, $(x_3, 0)$. In fact these roots and the leading coefficient determine

$$f(t) = -(x_2 - x_1)(x_3 - x_1)(x_3 - x_2) \prod_{i=1}^{3} (t - x_i).$$

Exercise 3.S.11: Geometric and structural determinant proofs

Use $p = (1, 4)$ as a base vertex, and take adjacent displacements $u = (-1, 5) - p = (-2, 1)$ and $v = (5, 8) - p = (4, 4)$. Their sum gives the remaining vertex: $p + u + v = (3, 9)$. This verifies the

parallelogram structure. Equivalently, the diagonals have the same midpoint because $(1,4)+(3,9) = (-1,5)+(5,8)$. Its area is

$$\left\| \begin{vmatrix} -2 & 4 \\ 1 & 4 \end{vmatrix} \right\| = |-8-4| = 12.$$

Exercise 3.S.12: Geometric and structural determinant proofs

A 2×2 matrix is invertible if and only if the parallelogram spanned by its two columns has positive area. Indeed, the columns are independent exactly when they are not parallel, which is exactly when the parallelogram has a nonzero base and positive perpendicular height. The Invertible Matrix Theorem equates that independence with invertibility. In determinant language the same area is $|\det A|$, but the geometric statement itself describes the nonflat image of the unit square.

Exercise 3.S.13: Geometric and structural determinant proofs

Put $d = \det A \neq 0$. The adjugate identity gives $A \operatorname{adj} A = \operatorname{adj} A\, A = dI$. Divide both equations by d:

$$(A/d) \operatorname{adj} A = I, \qquad \operatorname{adj} A\, (A/d) = I.$$

Thus A/d is a two-sided inverse of $\operatorname{adj} A$. This proves both its invertibility and $(\operatorname{adj} A)^{-1} = A/\det A$.

Exercise 3.S.14: Geometric and structural determinant proofs

A block-triangular determinant rule.

For $M = \left[\begin{smallmatrix} A & 0 \\ C & D \end{smallmatrix}\right]$, expand the determinant by its signed products, one entry from each row and column. Any nonzero product must use the first n columns for the top n rows, since the upper-right block is zero. Those top rows use *all* first n columns. The bottom n rows must therefore use the last n columns. Such a product splits into one signed product of A and one of D; the signs also multiply because no columns cross between blocks. Summing gives $\det M = (\det A)(\det D)$. This argument does not require either block to be invertible.

(a) Put $C = 0$ and $D = I_n$.

The rule gives $\det \left[\begin{smallmatrix} A & 0 \\ 0 & I \end{smallmatrix}\right] = (\det A)(\det I) = \det A$.

(b) Put the upper-left block equal to I_n.

Then $\det \left[\begin{smallmatrix} I & 0 \\ C & D \end{smallmatrix}\right] = (\det I)(\det D) = \det D$.

(c) Both block-triangular orientations.

The proved rule gives the lower-block-triangular formula. For the upper-block-triangular matrix, transpose:

$$\det \begin{bmatrix} A & B \\ 0 & D \end{bmatrix} = \det \begin{bmatrix} A^T & 0 \\ B^T & D^T \end{bmatrix} = (\det A^T)(\det D^T) = (\det A)(\det D).$$

The off-diagonal blocks affect entries but not the determinant in these zero-block arrangements.

Exercise 3.S.15: Geometric and structural determinant proofs

(a) Determine the factors by multiplying their blocks.

Block multiplication gives

$$\begin{bmatrix} I & 0 \\ X & I \end{bmatrix} \begin{bmatrix} A & B \\ 0 & Y \end{bmatrix} = \begin{bmatrix} A & B \\ XA & XB+Y \end{bmatrix}.$$

To match the lower-left block C, use $X = CA^{-1}$. To match the lower-right block D, use $Y = D - CA^{-1}B$. Multiplicativity and Exercise 14 now give

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = 1 \cdot (\det A) \det(D - CA^{-1}B).$$

The first factor has determinant 1 because its two diagonal blocks are identities.

(b) Use precisely the commuting hypothesis stated.

If $AC = CA$, then $ACA^{-1} = C$, and hence

$$A(D - CA^{-1}B) = AD - CB.$$

Taking determinants gives $(\det A)\det(D-CA^{-1}B) = \det(AD-CB)$. Substitution in part (a) proves the asserted formula. We have not commuted A with B or D; neither of those extra hypotheses is supplied or needed.

Exercise 3.S.16: Geometric and structural determinant proofs

Let $d = a - b$. The claimed determinant is $d^{n-1}[a + (n-1)b]$.

(a) Subtract successive rows in the correct order.

Perform $R_1 \leftarrow R_1 - R_2$, then $R_2 \leftarrow R_2 - R_3$, continuing downward through $R_{n-1} \leftarrow R_{n-1} - R_n$. At each step the row being subtracted is still its original row. The first $n-1$ rows now have d in position (i,i), $-d$ in position $(i,i+1)$, and zeros elsewhere. The last row remains $(b, b, \ldots, b, a)$. Every operation is a row replacement, so the determinant is unchanged.

(b) Add successive columns, now from left to right.

Apply $C_2 \leftarrow C_2 + C_1$, then $C_3 \leftarrow C_3 + C_2$ using the *new* column 2, and continue. The first $n-1$ rows become zero off their diagonal d entries. In the last row the entries accumulate to $b, 2b, \ldots, (n-1)b, a + (n-1)b$. Column replacements preserve the determinant, by the transpose version of the row rule.

(c) Read a triangular determinant.

The matrix is now lower triangular with diagonal $d, \ldots, d, a+(n-1)b$. Their product is $(a-b)^{n-1}[a+(n-1)b]$. No division by $a-b$ was used, so $a = b$ causes no gap in the proof. The case $n = 1$ simply says the determinant is a.

Exercise 3.S.17: Geometric and structural determinant proofs

What we must show. Reprove the formula by induction without assuming the preceding row-operation argument. For $n = 1$ the determinant is a. Suppose the formula holds at size $n - 1$. By linearity in column 1, with all other columns fixed, $\det A = \det B + \det C$ for the displayed matrices.

In B, expand down column 1: only the first entry $a - b$ is nonzero. The remaining $(n - 1) \times (n - 1)$ matrix has diagonal a and off-diagonal entries b. The induction hypothesis therefore gives

$$\det B = (a - b)^{n-1}[a + (n - 2)b].$$

In C, subtract row 1 from every other row. Column 1 then has b followed by zeros, and the remaining lower-right block is $(a - b)I_{n-1}$. Thus $\det C = b(a - b)^{n-1}$. Adding these two expressions yields

$$\det A = (a - b)^{n-1}[a + (n - 2)b + b] = (a - b)^{n-1}[a + (n - 1)b].$$

This proves the induction step, including zero parameter values because no division occurs.

Exercise 3.S.18: Two constant-off-diagonal examples

For the first matrix, $n = 4$, $a = 3$, $b = 8$. The formula gives

$$(3 - 8)^3[3 + 3 \cdot 8] = (-5)^3(27) = -3375.$$

For the second, $n = 5$, $a = 8$, $b = 3$, so

$$(8 - 3)^4[8 + 4 \cdot 3] = 5^4(20) = 12500.$$

An independent exact computation is

```
from sympy import Matrix
for n,a,b in ((4,3,8),(5,8,3)):
    A = Matrix(n,n,lambda i,j: a if i == j else b)
    print(A.det())
```

It prints -3375 and 12500, respectively. A negative determinant in the first example describes orientation, not a negative ordinary volume.

Exercise 3.S.19: A matrix of successive minima

Let H_n have entry $\min(i, j)$ in position (i, j). The three displayed examples are H_3, H_4, H_5. Exact computation gives determinant 1 for each:

```
from sympy import Matrix
for n in (3,4,5):
    H = Matrix(n,n,lambda i,j: min(i+1,j+1))
    print(H.det())
```

Prove the conjecture for every size.

Starting at the *bottom*, do $R_n \leftarrow R_n - R_{n-1}$, then $R_{n-1} \leftarrow R_{n-1} - R_{n-2}$, and continue to $R_2 \leftarrow R_2 - R_1$. This order ensures that every subtracted row is still its original row. Row 1 is all 1's; new row i has $i - 1$ zeros followed by 1's. Thus the resulting matrix is upper triangular with every diagonal entry 1. No operation changed the determinant, so $\det H_n = 1$ for every $n \geq 1$.

Exercise 3.S.20: A related matrix with a different first increment

What we must show. Determine the determinant of the matrix exactly as printed. Its successive distinct entries are $d_1 = 1$ and $d_i = 3(i-1)$ for $i \geq 2$: in particular $1, 3, 6, 9, \ldots, 3(n-1)$. These are *not* the triangular numbers after the first three entries. The (i, j) entry is $d_{\min(i,j)}$.

Run the same bottom-up differences.

For $i = n, n-1, \ldots, 2$, replace R_i by $R_i - R_{i-1}$. The resulting matrix is upper triangular with diagonal

$$d_1, \quad d_2 - d_1, \quad d_3 - d_2, \ldots, d_n - d_{n-1} = 1, 2, 3, \ldots, 3.$$

Therefore

$$\boxed{\det A = 2 \cdot 3^{n-2} \quad (n \geq 2).}$$

At $n = 1$, the matrix is [1] and has determinant 1. A direct check for $n = 2, 3, 4, 5$ gives $2, 6, 18, 54$:

```
from sympy import Matrix
for n in (2,3,4,5):
    d = [1] + [3*i for i in range(1,n)]
    A = Matrix(n,n,lambda i,j: d[min(i,j)])
    print(A.det())
```

The special first difference $3 - 1 = 2$ is why a guess of 3^{n-1} would be wrong. Equivalently, the triangular matrix after row differences has a leading 2×2 block of determinant 2 and a remaining block equal to 3 times the upper-triangular matrix from Exercise 19; block-triangular determinants give the same answer.